

5SEE – MACHINE LEARNING FOR ELECTRONIC DESIGN

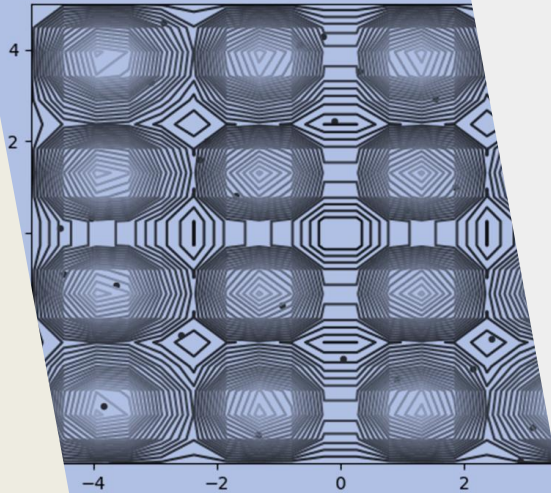
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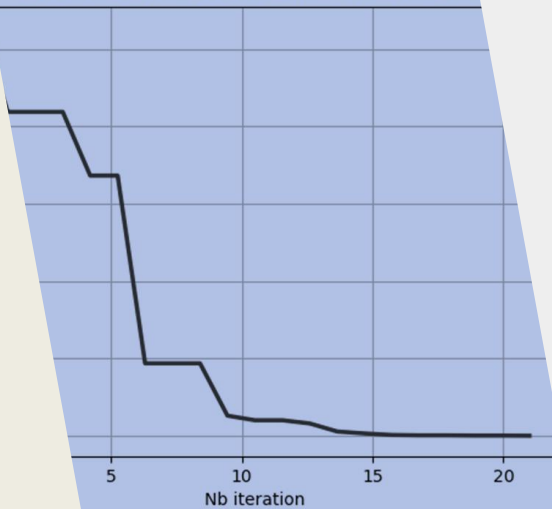
November 2025

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iter = 0



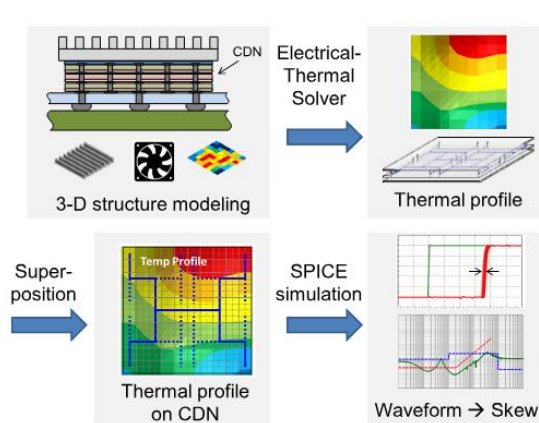
Evolution of best solution after each interatio



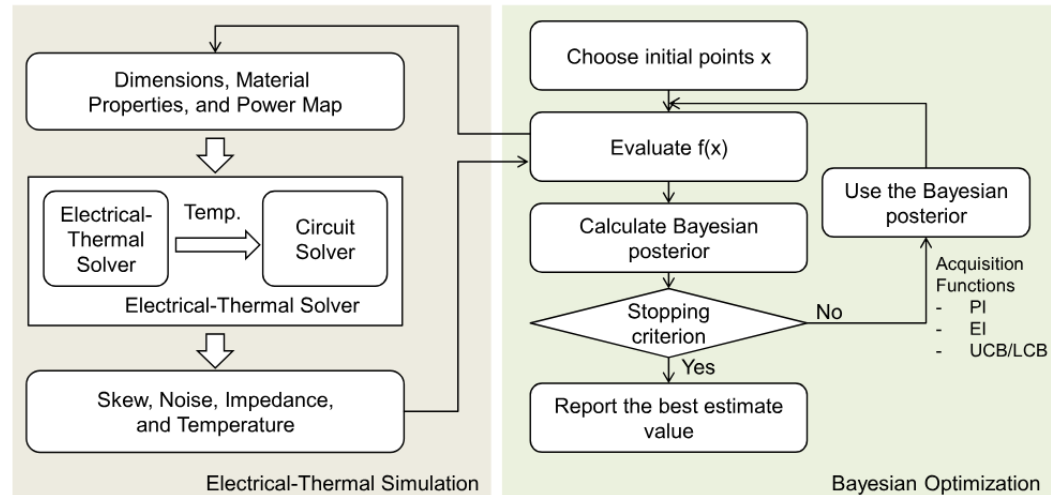
Introduction

AI/ML usage in electronic design ? Some examples

- ✓ Electro-thermal simulation to optimize clock delivery network in 3D IC module (clock skew and thermal gradient)

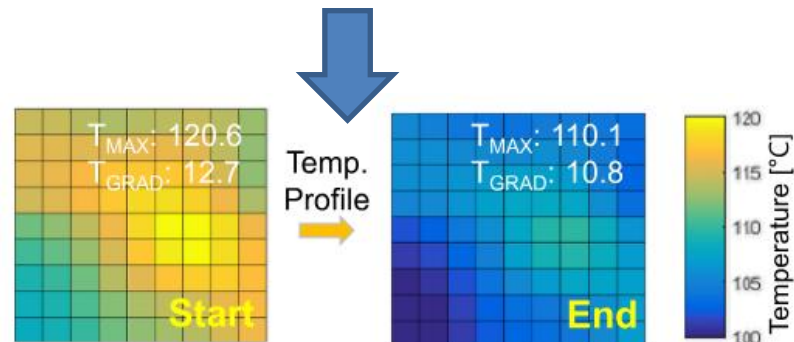


Expensive simulation



Bayesian-driven flow for electrical-thermal optimization

S. J. Park, B. Bae, J. Kim, M. Swaminathan, "Application of Machine Learning for Optimization of 3-D Integrated Circuits and Systems," IEEE Trans. On VLSI Systems, vol. 25, no. 6, June 2017.

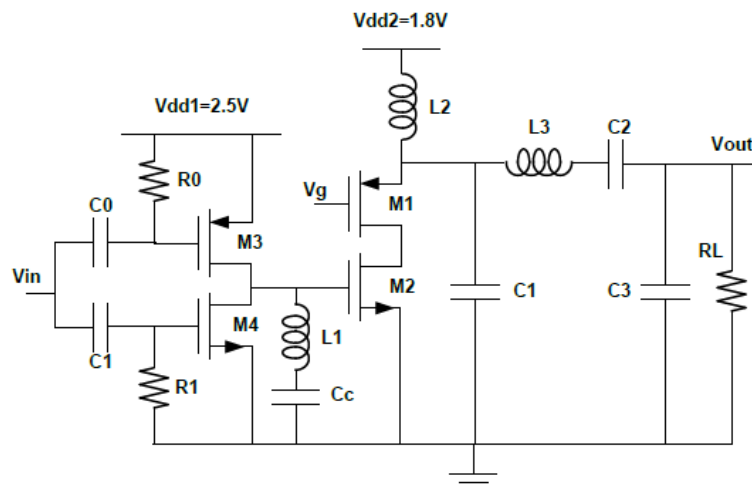


Improvement of temperature distribution on die surface

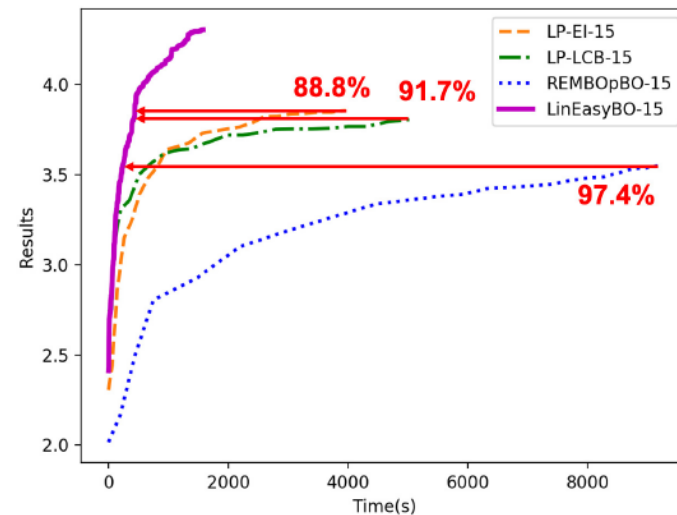
Introduction

▶ AI/ML usage in electronic design ? Some examples

- ✓ Use of machine-learning method to accelerate analog circuit sizing.
- ✓ Example: use of Bayesian optimization to optimize the output power and efficiency of a class-E RF power amplifier without human intervention.



12 design variables



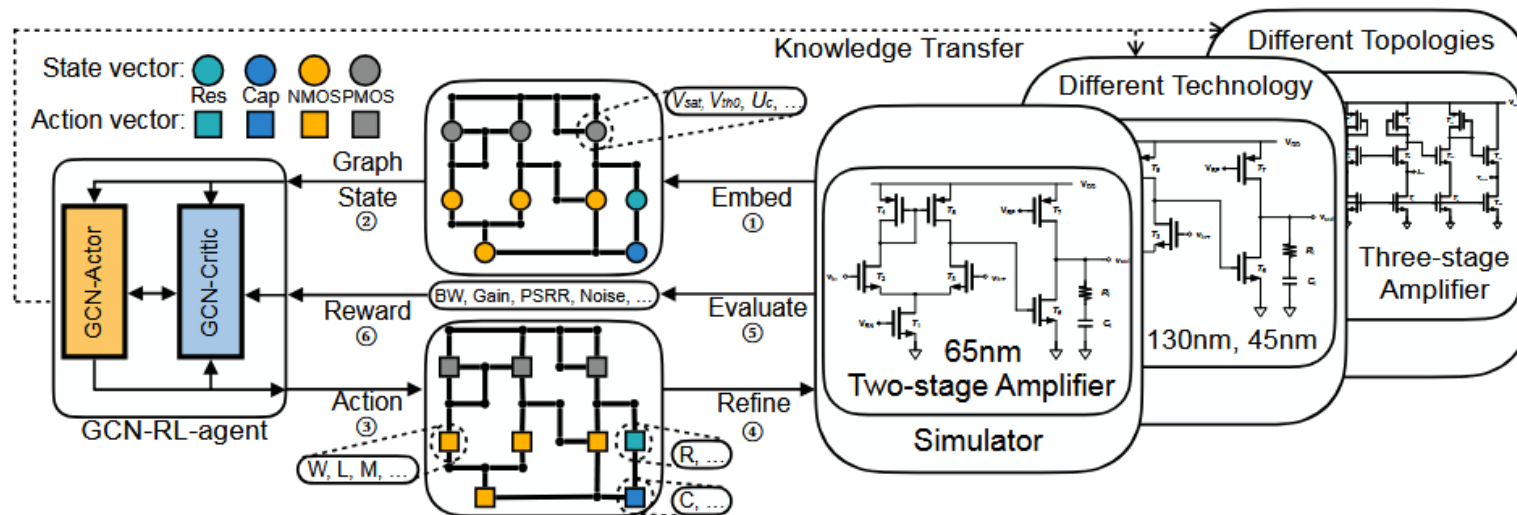
LP-EI-15	4.34	3.49	3.86	0.23	1h6m3s
LP-LCB-15	4.08	3.41	3.81	0.15	1h23m40s
REMBOpBO-15	4.10	2.92	3.55	0.22	2h33m23s
LinEasyBO-15	5.96	3.21	4.30	0.74	27m23s

S. Zhang, F. Yang, C. Yan, D. Zhou, X. Zeng, "LinEasyBO: Scalable Bayesian Optimization Approach for Analog Circuit Synthesis via One-Dimensional Subspaces," 4th Workshop on Machine Learning for CAD (MLCAD), Sept. 2022.

Introduction

► AI/ML usage in electronic design ? Some examples

- ✓ Automatic transistor sizing for analog circuits when transfer to another technological nodes ?
- ✓ Limit of manual design to ensure similar figures-of-merit (intensive and time-consuming approach due to large design space, slow simulation tool and sophisticated trade-off)



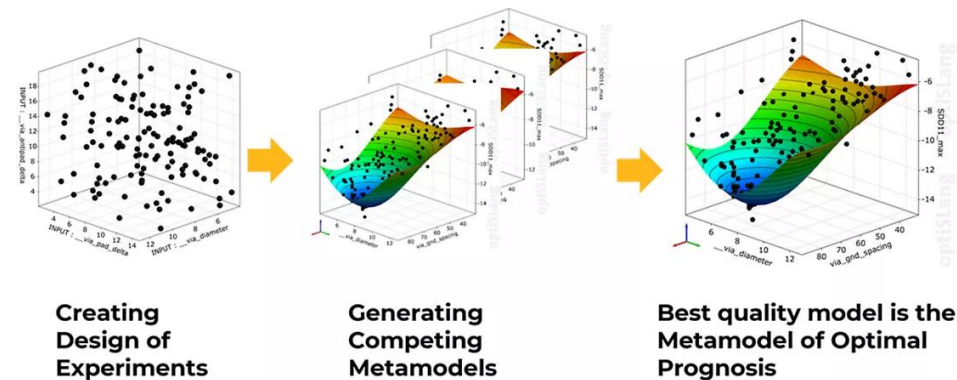
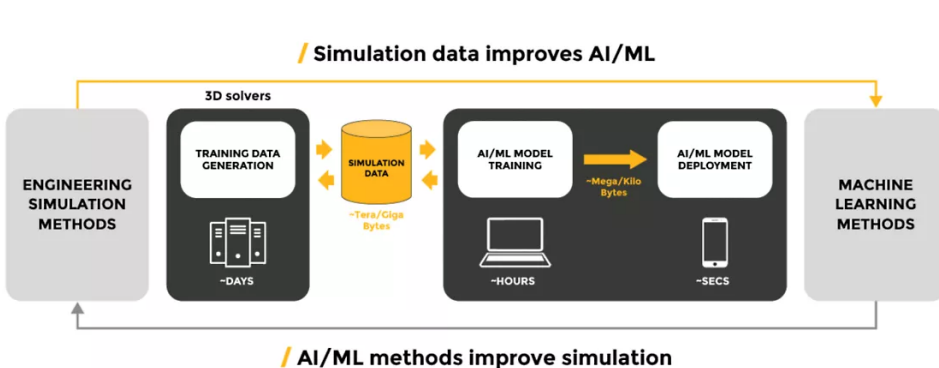
Graph Convolutional Neural Networks based Reinforcement Learning for Automatic Transistor Sizing

H. Wang, K. Wang, J. Yang, L. Shen, N. Sun, H. S. Lee, S. Han, "GCN-RL Circuit Designer: Transferable Transistor Sizing with Graph Neural Networks and Reinforcement Learning," IEEE Design Automation Conference, 2020.

Introduction

AI/ML usage in electronic design ? Some examples

- ✓ Integration of artificial intelligence (AI) and machine learning (ML) methods in commercial EDA simulation tools to enhance design exploration and optimization, accelerate simulations.



<https://www.ansys.com/fr-fr/blog/optimize-design-simulation-with-ai-ml-metamodeling>

Purpose of this course:

- ✓ What are the on-going trends?
- ✓ What are the typical AI/ML algorithms to accelerate the electronic design process?
- ✓ Experiment AI/ML algorithms for a typical electronic design optimization (coupling an electronic design tool to an AI/ML Python library)

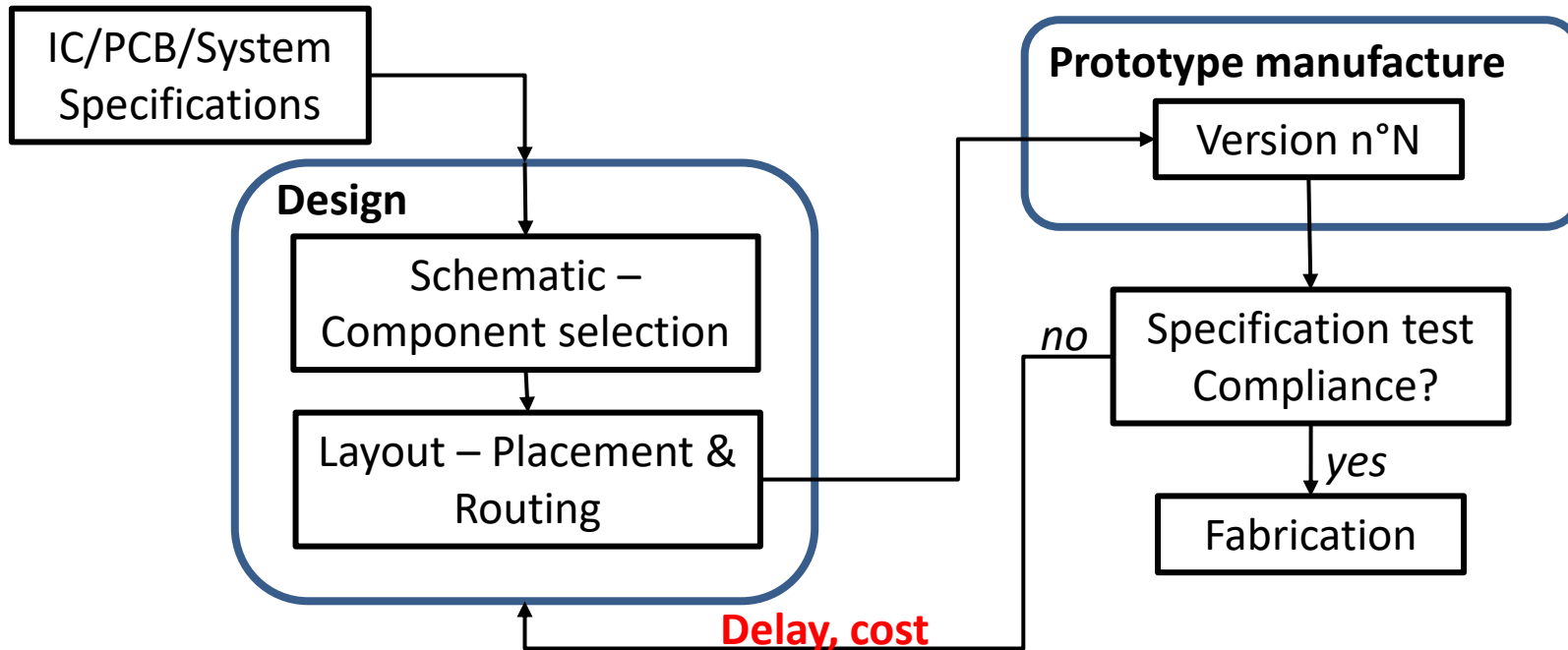
Summary

- 1. Overview of Electronic Design Automation (EDA) Tools and Usage**
- 2. Current challenges of EDA – Benefits of AI/ML for EDA**
- 3. Overview of AI/ML methods**
- 4. Metaheuristic for electronic design optimization**
- 5. Supervised learning methods for surrogate modeling**
- 6. Reinforcement learning for electronic design optimization**

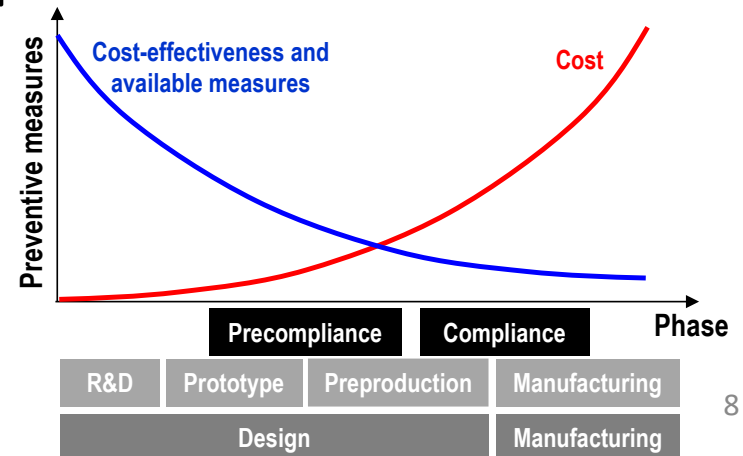
1. Overview of Electronic Design Automation (EDA) Tools and Usage

Overview of EDA Tools and Usage

► Electronic Design Flow – Trial-and-error design flow

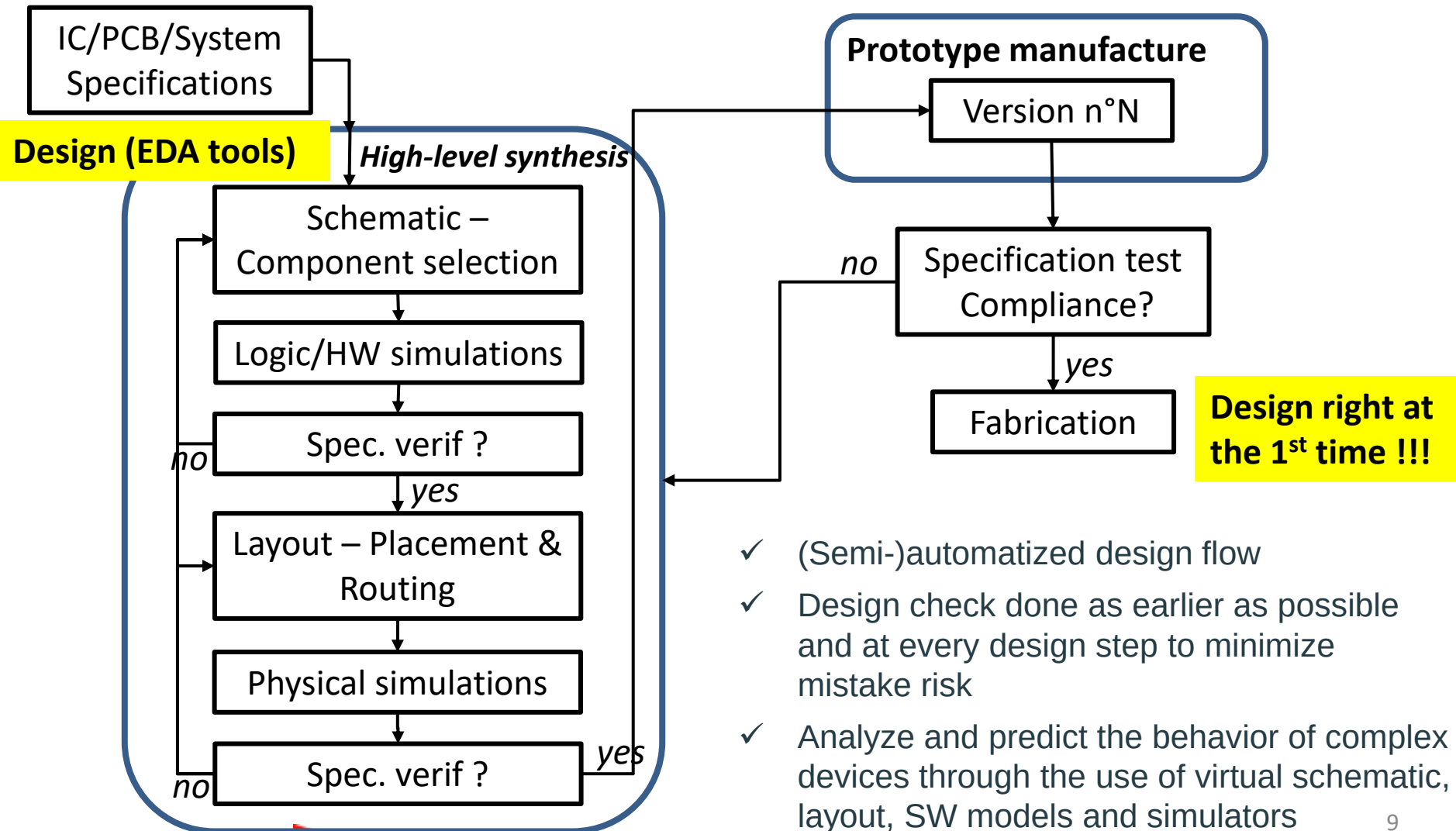


Evolution of cost of design changes during the design phase:



Overview of EDA Tools and Usage

► Electronic Design Flow – EDA-based design flow



Overview of EDA Tools and Usage

► Simulation and models for electronic design

Technological data (confid.)

Design variables

Carrier drift-diffusion equation $J = q(n\mu E + D\nabla n)$

Kirchoff's laws $\sum_k I_k = 0$ $\sum_i E_i - \sum_n Z_n I_n = 0$

⋮

**Modelling and
simulations**

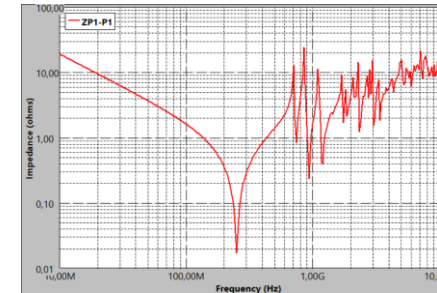
Maxwell's equations $\nabla \times E = -\frac{\partial B}{\partial t}$

Thermal diffusion equation $\rho C_P \frac{\partial T}{\partial t} + \nabla \cdot j = S$

⋮

**Response
prediction**

**Functional ? Manufacturable ?
Robust ? Reliable ?**



- ✓ Design validation/optimization of electronic devices/systems requires physical modeling and simulation when the complexity increases.
- ✓ Main issues:
 - ✓ Non-linear and multiphysical models → numerical solvers (computationally intensive)
 - ✓ Response related to proprietary technological data
 - ✓ Prediction hardly based on simplified model or generalized response.

Overview of EDA Tools and Usage

► EDA – Electronic Design Automation

EDA = software and specialized hardware tools for the design, verification, manufacturing and testing of electronic systems (ICs, PCBs, FPGAs, ...)

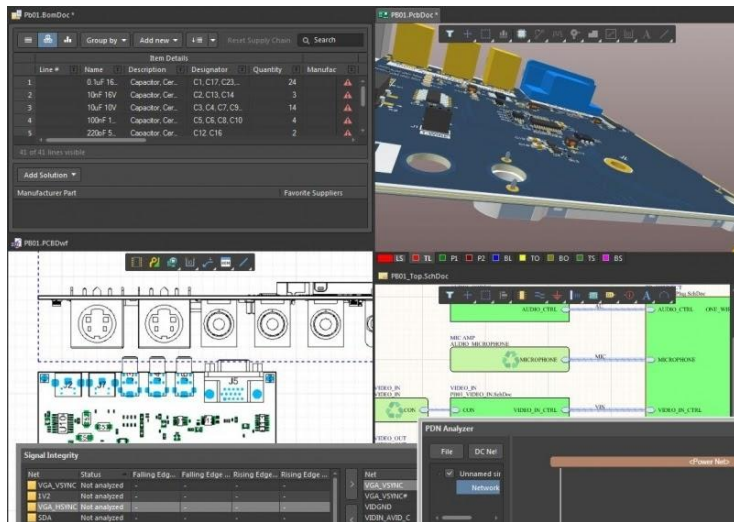
- ✓ Developed since the 1980's (initially for IC design)
- ✓ Essential when electronic devices are too complex to design and responses unpredictable without the help of computers
- ✓ EDA tools integrate several CAD interfaces and numerical simulation tools to predict, optimize, verify and analyze the performances and the manufacturability.

Overview of EDA Tools and Usage

EDA vs. CAD (Computer-Aided Design) tools

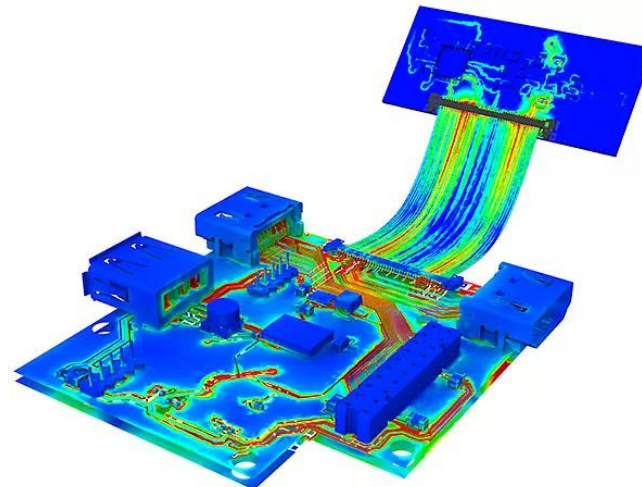
- ✓ CAD: 2D/3D modeling and visualization. Possible link with simulations
- ✓ EDA: SW dedicated for the different steps of electronic system design (schematic, simulation layout design, verification, ...)

CAD tool



(Altium Designer)

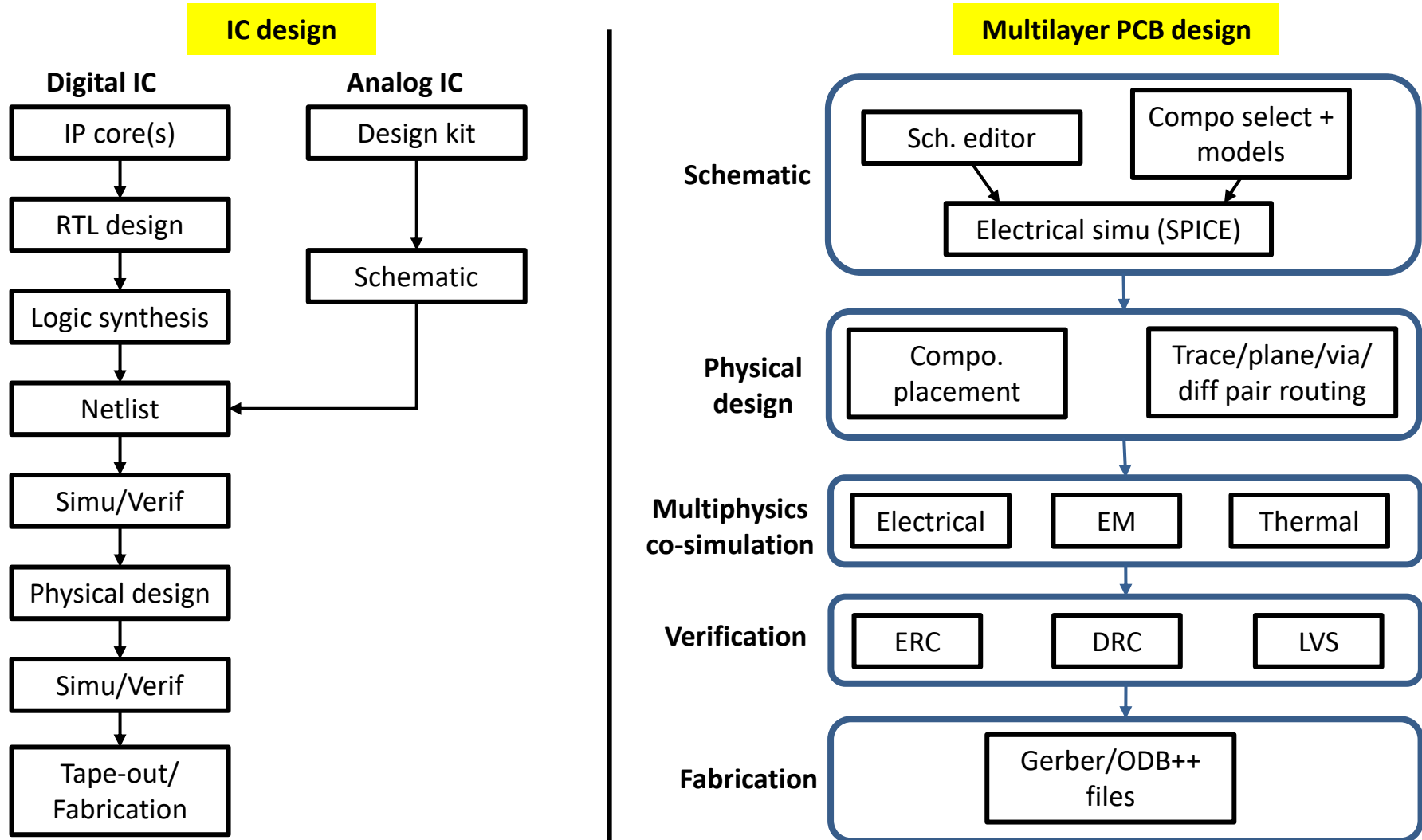
EDA tool



(Ansys HFSS: Electromagnetic simulation of PCBs)

Overview of EDA Tools and Usage

EDA in IC and PCB design – Typical flow

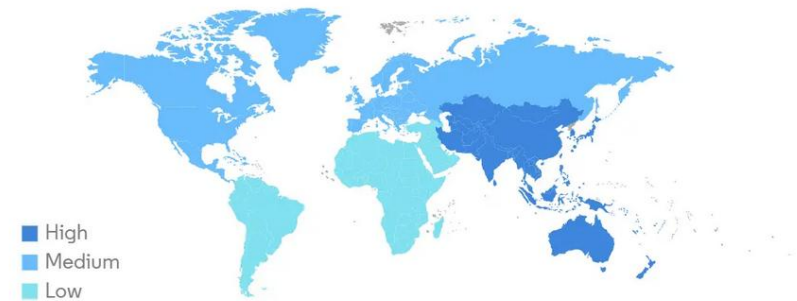


Overview of EDA Tools and Usage

EDA market

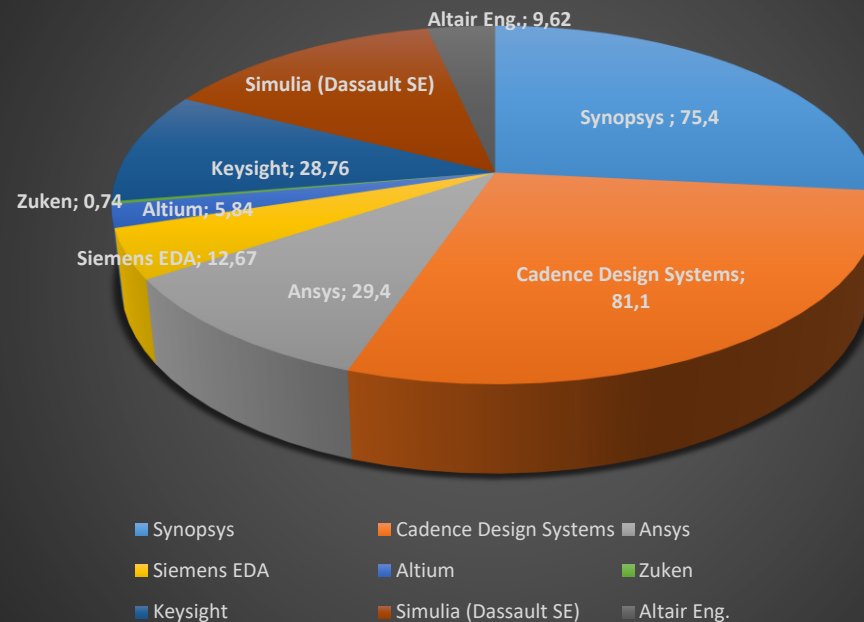
- ✓ Market of 19.22 Billions of Dollars in 2025
- ✓ Growth rate of +8.5 % (2025-2030)
- ✓ Main EDA vendors and market capitalization (Billions of \$ in 2025):

Global Electronic Design Automation Tools (EDA) Market CAGR (%), Growth Rate by Region, 2025 - 2030



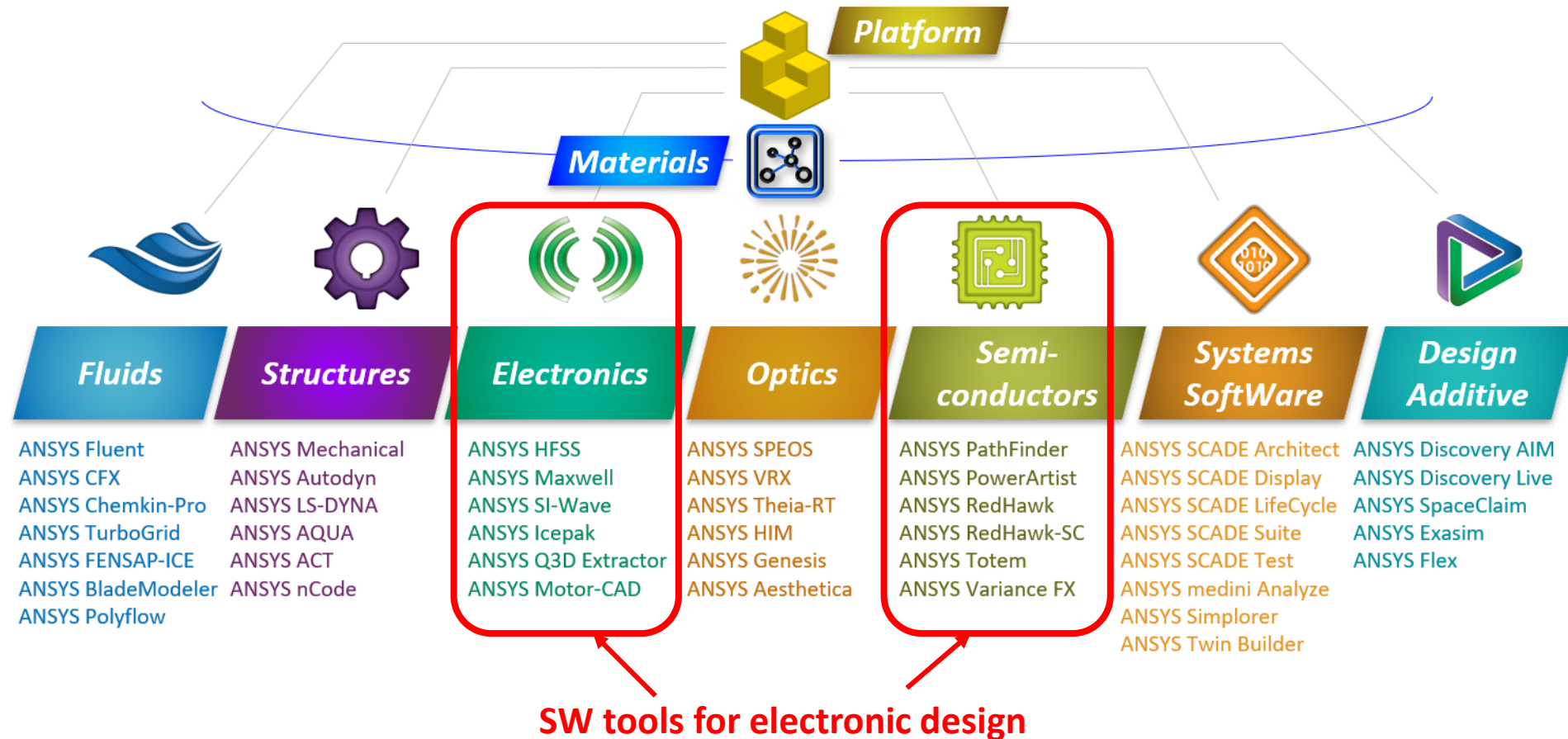
Source: Mordor Intelligence

EDA market capitalization in 2025 - Main EDA vendors



Overview of EDA Tools and Usage

EDA software: an example - ANSYS



2. Current challenges of EDA – Benefits of AI/ML for EDA

Current challenges of EDA – Benefits of AI/ML for EDA

► Current challenges for EDA

- ✓ Computational time and resources for some simulations (e.g. SPICE simu. with transistor-based models, 3D EM/thermal simulations, ...)
- ✓ Simulation data are usually sparse
- ✓ Multi-objective design optimization needs to adjust tens of design variables → limit of human intervention in the optimization process
- ✓ Problem of confidentiality when IP cores and models are shared

- ✓ AI/ML → new methodologies and tools for electronic designers to solve more sophisticated problems
- ✓ Main purpose of AI/ML in EDA: automatic processing of simulation data for:
 - Surrogate modeling (data-driven model) → accelerate the simulation and IP preservation
 - Fully automatized design under optimization constraints, exploration of new design solutions
 - Sensitivity analysis and uncertainty propagation

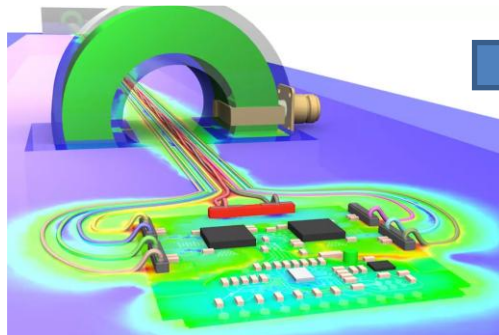
Current challenges of EDA – Benefits of AI/ML for EDA

► Traditional physics modeling vs. Surrogate modeling

- ✓ Surrogate modeling (or metamodeling) = generation of a simplified and computationally-efficient approximation of a more complex, high-order model.
- ✓ Example: EM/SI/PI analysis of a PCB:

Traditional numerical method

Numerical solving of
diff. equations (e.g.
FEM, FDTD, MoM,
TLM, ...)

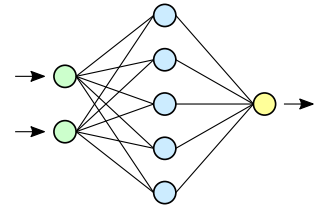


(Ansys – simulation of BCI on
automotive harness)

Hours/days of simulation to predict
PCB responses for different
combinations of design variables

Surrogate modeling

Training model to
approximate physical
behavior of all system or a
part of the system



Seconds/ minutes of simulation to
predict PCB responses for different
combinations of design variables

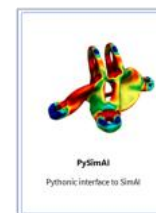
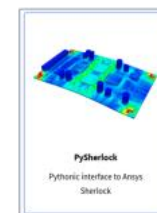
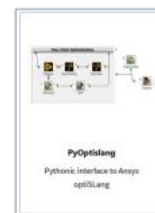
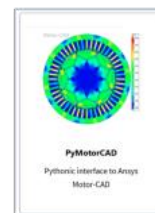
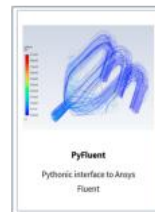
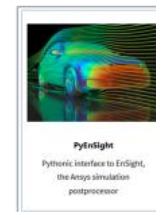
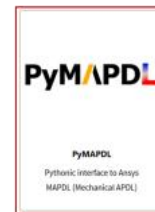
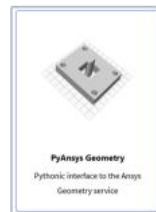
Current challenges of EDA – Benefits of AI/ML for EDA

► Python automation for AI/ML

- ✓ Scripts are used for a long time to control and interface with EDA tools
- ✓ Problem: proprietary scripting language
- ✓ Things change for 10 years with the general development of Python interfacing and automatizing for most EDA tools.
- ✓ Advantages:
 - Popular language and many open source code and libraries (especially for data processing, AI and ML)
 - Flexible and automatic deployment of design process, without human input
 - Launch automatic process of data generation and ML models



Python + ANSYS = Py/Ansys



« An Introduction to PyAnsys:
Revolutionizing Engineering
Simulations », Mar 2025, Ansys.

3. Overview of AI/ML methods

Overview of AI/ML methods

► Overview

Artificial Intelligence

Machine Learning

Natural
Language
Processing (NLP)

Reasoning

Planning

Knowledge
Representation
and Reasoning

Vision

Motion and
Manipulation

Multiagent
Systems

General
Intelligence

Supervised
Learning

Semi-supervised
Learning

Unsupervised
Learning

Reinforcement
Learning

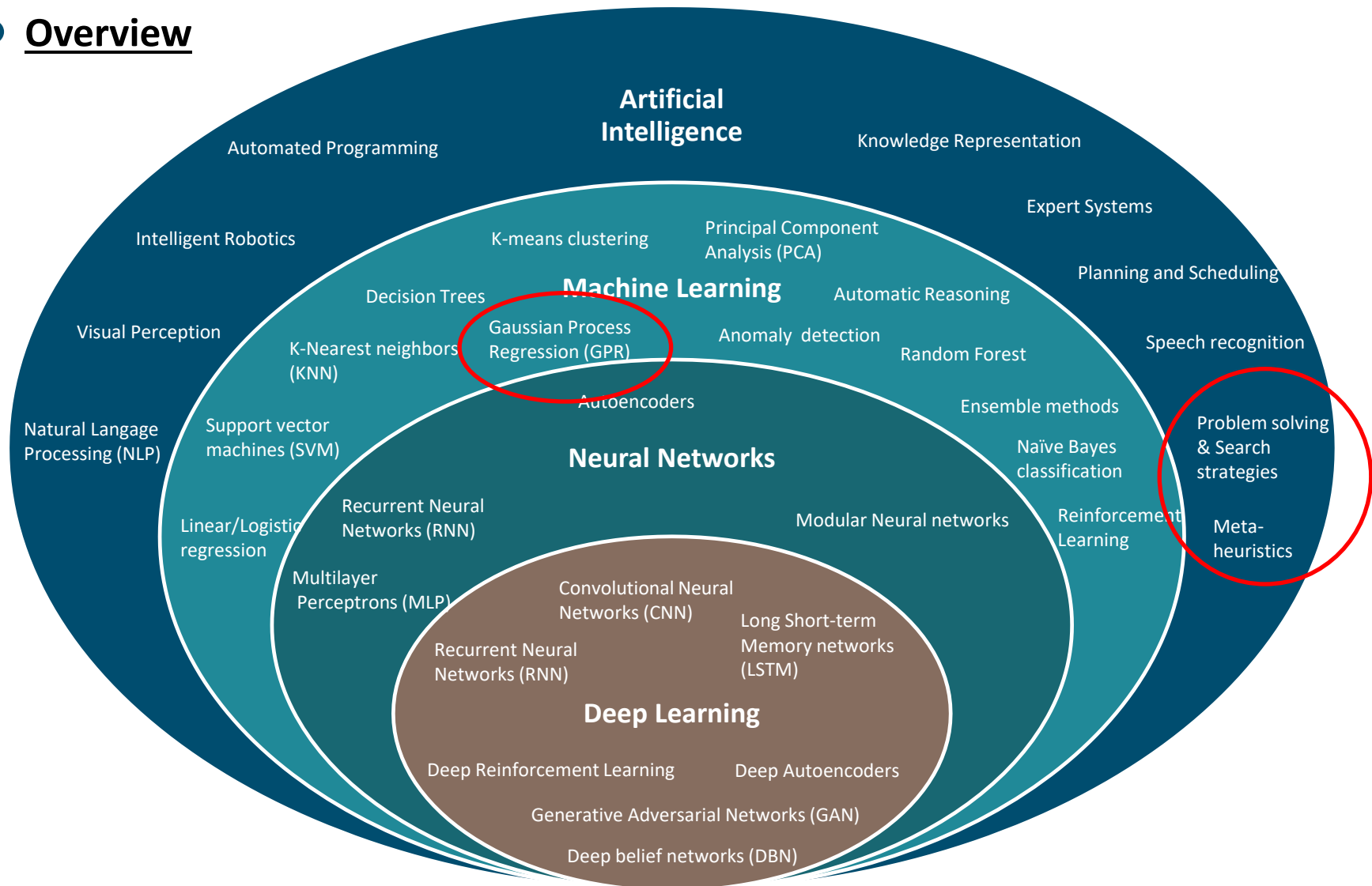
Transfer
Learning

Deep Learning

Course I4AEML21 – Machine Learning (A. Bit-Monnot, E. Chanthery, M-J. Huguet, P. Leleux, M. Siala)

Overview of AI/ML methods

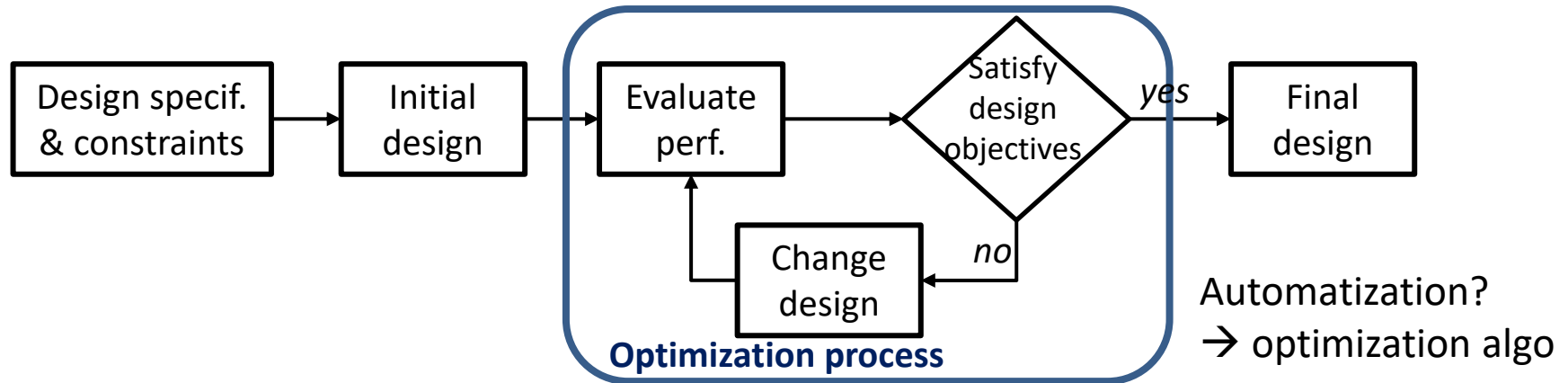
► Overview



4. Metaheuristics for electronic design optimization

Metaheuristics for electronic design optimization

► Typical engineering optimization process



✓ Advantages:

- Reduce the risk of human errors, avoid misleading intuition
- Accelerate design process when design variables increase (humans manage 2-3 variables)
- Systematic procedure

✓ Drawbacks

- High computational resources and time
- Depend on problem specified by designer (algo may find inadequate solutions)
- Optimal design may be counterintuitive and difficult to interpret
- Most of optimization algo do not guarantee to find the optimal design (local vs. global minimum)

Metaheuristics for electronic design optimization

► Optimization problem

Objective or cost function
(explicitly known or
black-box)

Design point

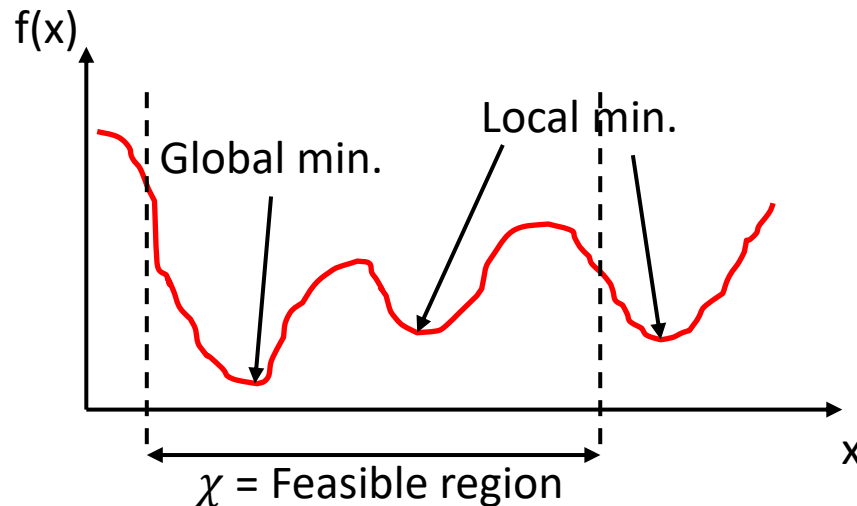
N design variables (real
or integer values)

$$\arg \min_x f(x), x = [x_0; x_1; \dots; x_N]$$

subject to $x \in \chi$

Constraints (linear
and/or non-linear)

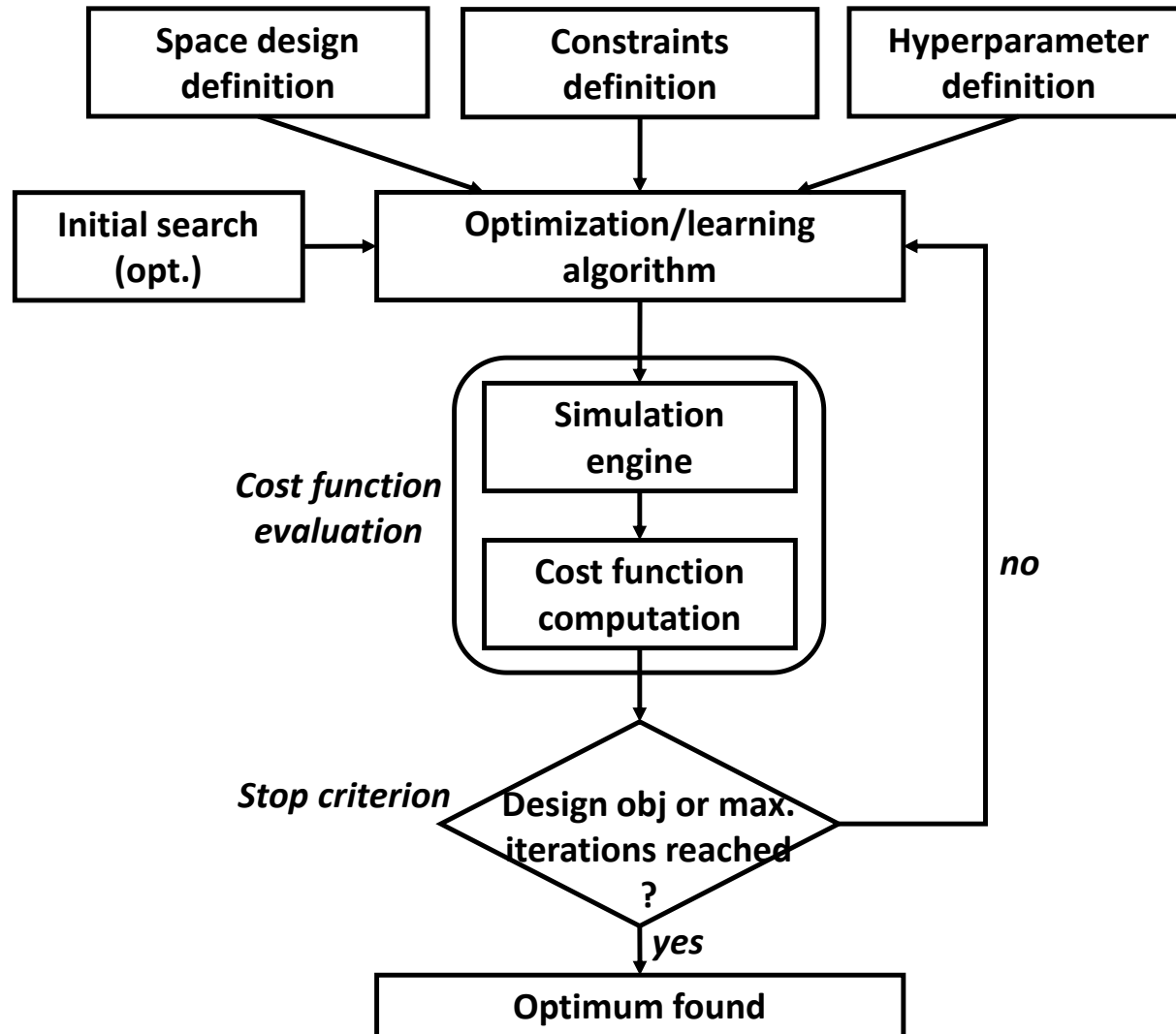
✓ 1D problem illustration:



Purpose: if it exists, find the global minimum in the feasible region defined by the constraints

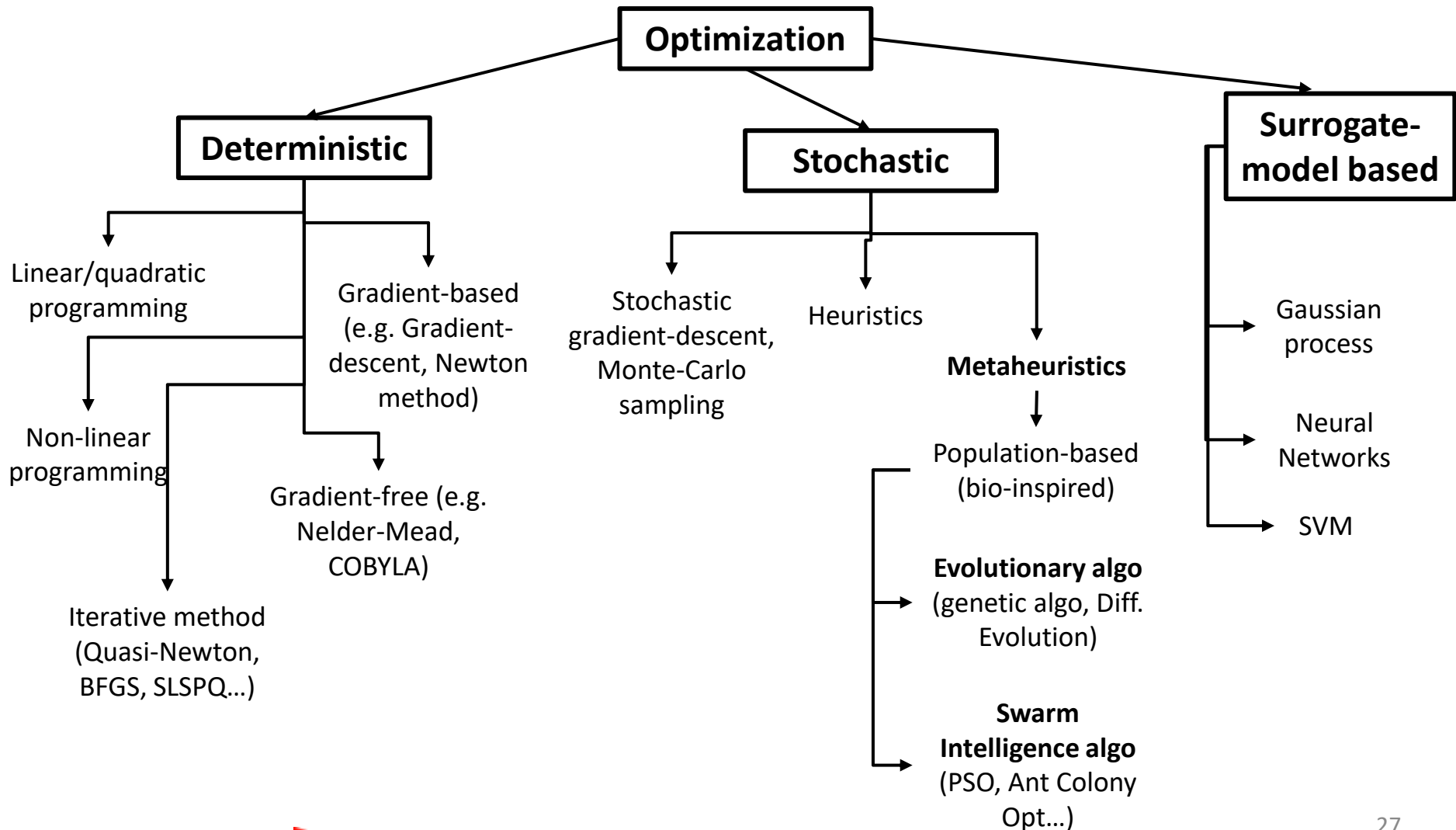
Metaheuristics for electronic design optimization

► Typical optimization flow in electronic design



Metaheuristics for electronic design optimization

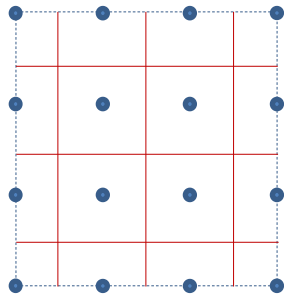
► Overview of optimization algorithms (not exhaustive list)



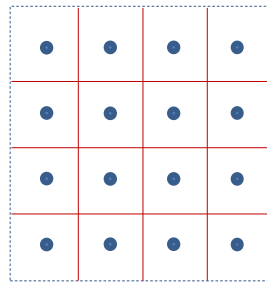
Metaheuristics for electronic design optimization

► Design space sampling

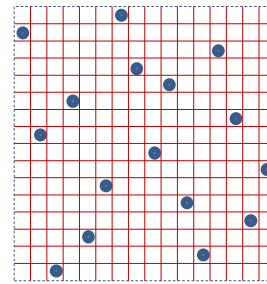
- ✓ Initial search requires an initial design space sampling.



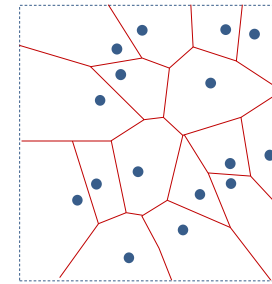
Regular grid



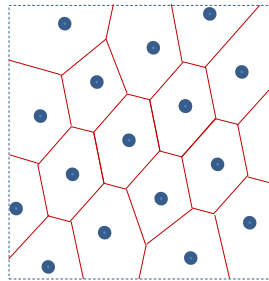
Sukharev grid



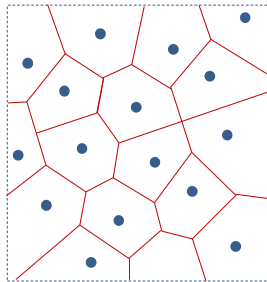
Latin Hypercube sampling



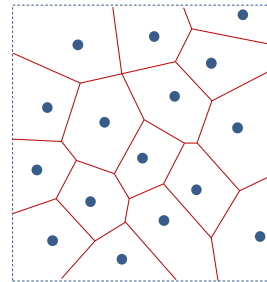
pseudo-random sampling



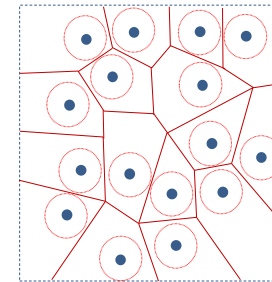
Lattice sampling



Hammersley sampling



Halton sampling

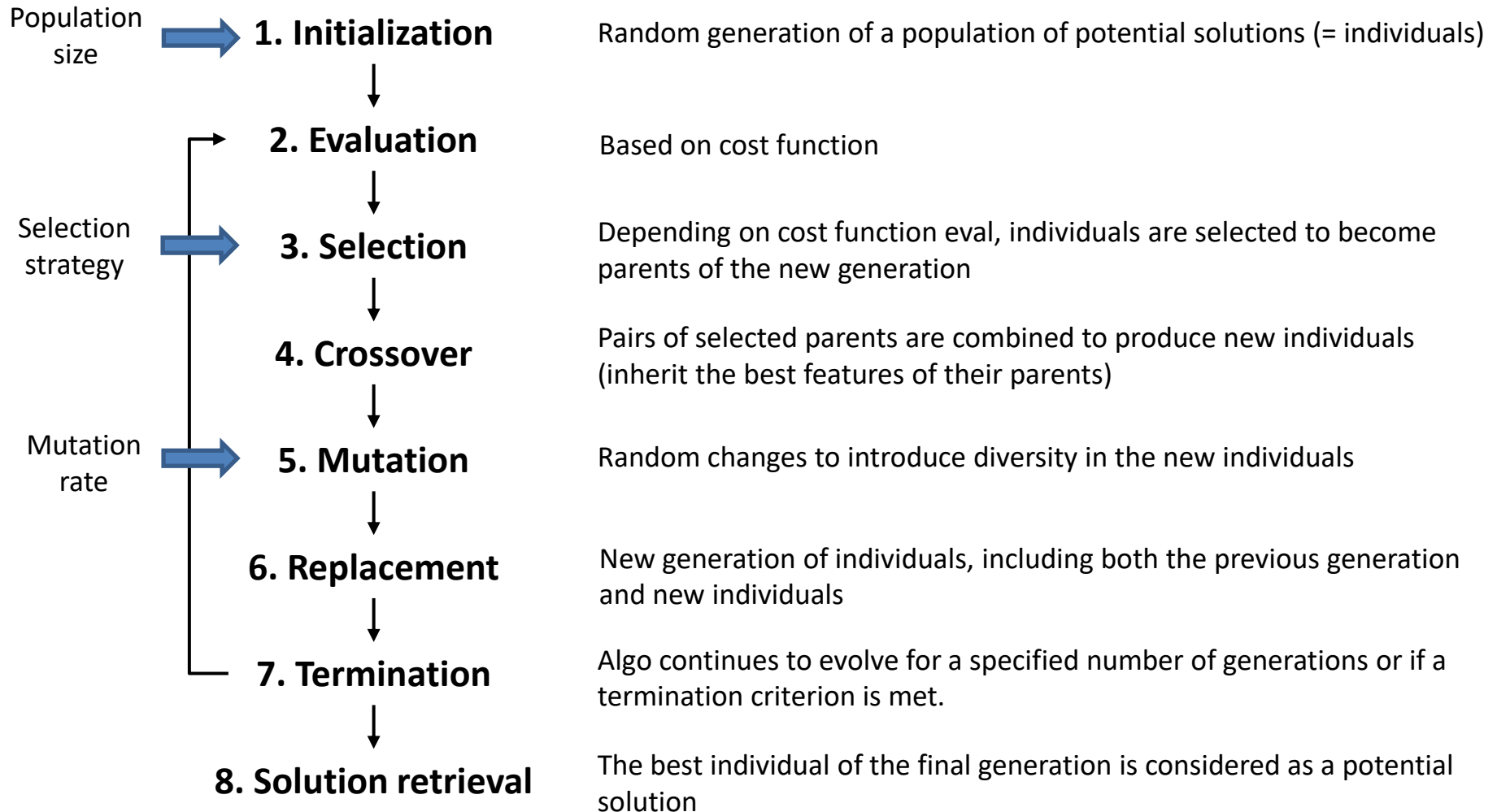


Poisson disk sampling

S. Serpaud et al., "Efficiency of Sequential Spatial Adaptive Sampling Algorithm to Accelerate Multifrequency Near-Field Scanning Measurement", IEEE Trans. on EMC, 2022.

Metaheuristics for electronic design optimization

► Genetic algorithm



Metaheuristics for electronic design optimization

► Differential evolution

- ✓ Simplified approach of genetic algorithm, usually more performant. Heavily parallelized.

Population size N → **1. Initialization**

Random generation of N individuals: individual $x = [x_0; x_1; \dots; x_n]$

2. Evaluation

Based on cost function f

Impact of N, F and CR ?

Evolution strategy, F →

3. Evolution

$X' = x_{\text{best}} + F \times (x_{\text{rand0}} - x_{\text{rand1}})$, F = mutation coefficient $\in [0; 2]$

Crossover proba CR →

4. Crossover

Random selection of a number $r_i \in [0; 1]$

For each entry i of an individual :
$$x_{\text{new}}[i] = \begin{cases} x'[i] & \text{if } r_i < CR \\ x_{\text{old}}[i] & \text{if } r_i > CR \end{cases}$$

5. Evaluation

Cost function evaluation of new individuals.

If $f(x_{\text{new}}) < f(x_{\text{old}})$, x_{new} replaces x_{old}

7. Termination

Algo continues to evolve for a specified number of generations or if a termination criterion is met.

8. Solution retrieval

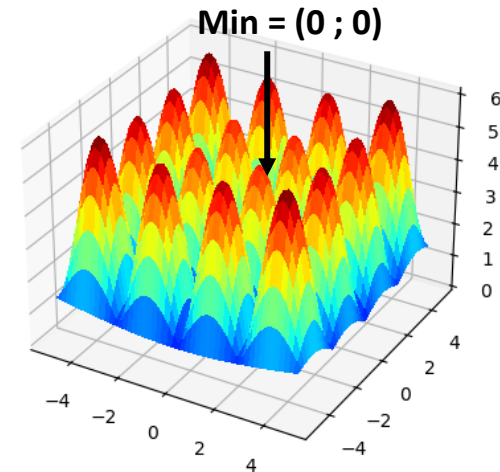
The best individual of the final generation is considered as a potential solution

Metaheuristics for electronic design optimization

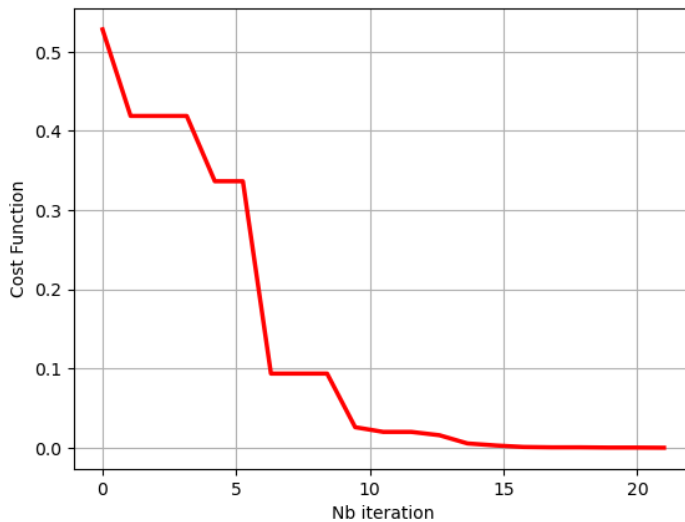
► Differential evolution

✓ Example: minimize $f(x, y) = \sqrt{x^2 + y^2} + \sin\left(\frac{2\pi}{5}x\right) \sin\left(\frac{2\pi}{5}y\right)$.

- Bounds = $[-5; 5]$, $[-5; 5]$
- Population size = 40
- Nb of iterations = 20
- Proba mutation = 0.7
- Coef mutation = 0.5



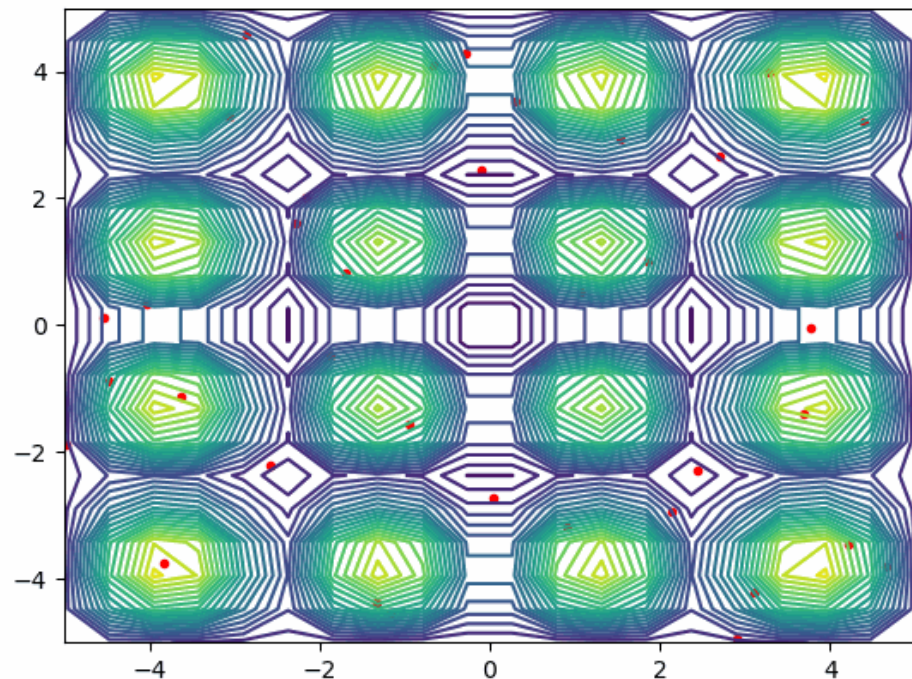
Evolution of best solution after each iteration



Convergence

Solution found = (4.440e-04 ; -2.441e-04)

iter = 0



Metaheuristics for electronic design optimization

► Particle Swarm Optimization

- ✓ Based on the movement of individual particles. Heavily parallelized.

Population
size N



1. Initialization

Random selection in the design space of speed $P_i(0)$ of N particles.
Random selection of initial speed $V_i(0)$

$t = 0$

2. Evaluation

Based on cost function f . P_{best_i} = Best position of particle $n^{\circ}i$.
 g_{best} = Best position of the swarm

Inertia weight
 w , cognitive
param $c1$,
social param
 $c2$



3. Speed update

Random selection of a number $r_i \in [0; 1]$
 $V_i(t+1) = w \cdot V_i(t) + C_1 r_1 \cdot (P_{best_i} - P_i(t)) + C_2 r_2 \cdot (g_{best} - P_i(t))$

4. Position update

$P_i(t+1) = P_i(t) + V_i(t+1)$

5. Evaluation

Based on cost function f . P_{best_i} = Best position of particle $n^{\circ}i$.
 g_{best} = Best position of the swarm

7. Termination

Algo continues to evolve for a specified number of iterations or if a termination criterion is met.

8. Solution retrieval

g_{best} is considered as a potential solution.

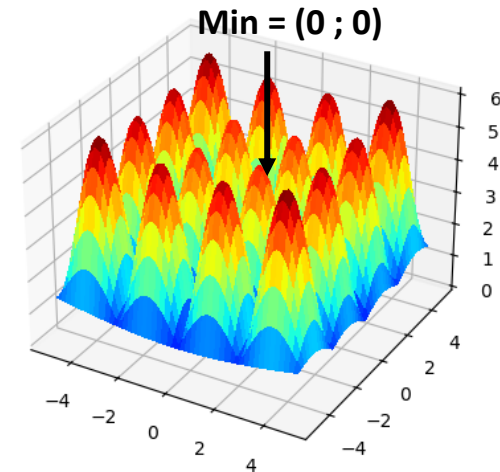
**Impact of N, w,
C1 and C2 ?**

Metaheuristics for electronic design optimization

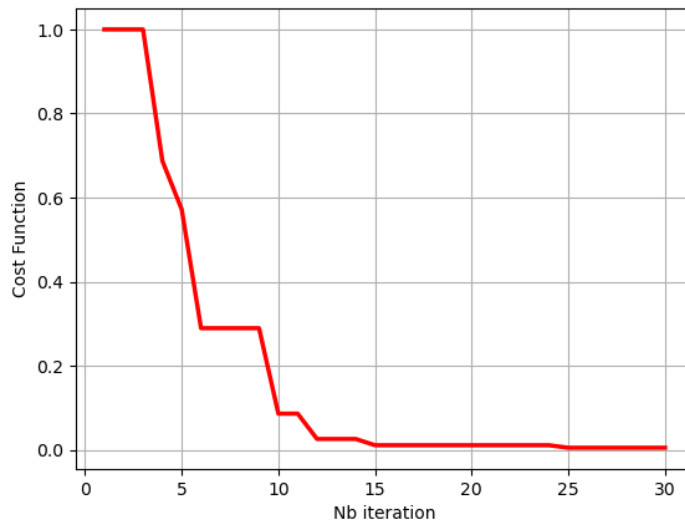
► Particle Swarm Optimization

✓ Example: minimize $f(x, y) = \sqrt{x^2 + y^2} + \sin\left(\frac{2\pi}{5}x\right) \sin\left(\frac{2\pi}{5}y\right)$.

- Bounds = $([-5;5], [-5;5])$
- Population size = 30
- Nb of iterations = 30
- $W = 0.8$
- $C1 = C2 = 0.5$



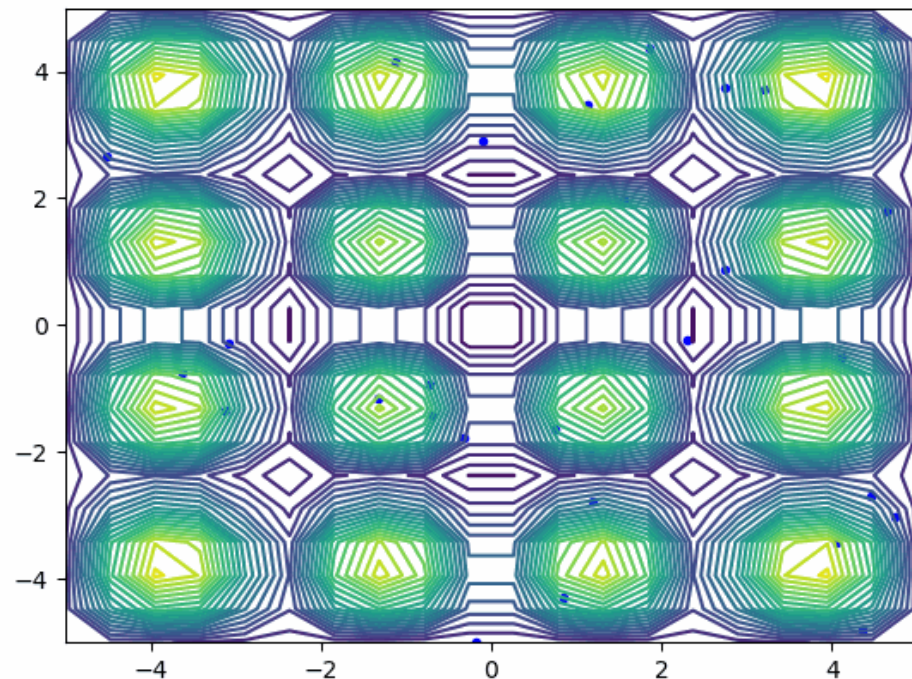
Evolution of best solution after each iteration



Convergence

Solution found = (0.01429838 ; -0.01161504)

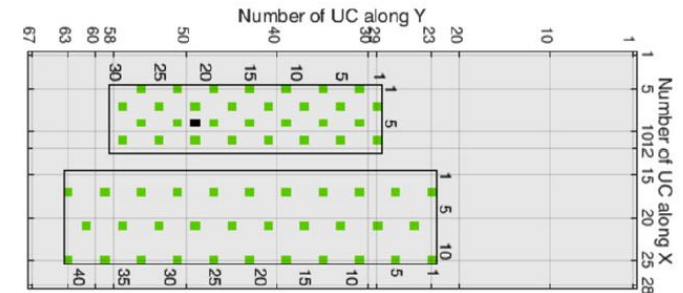
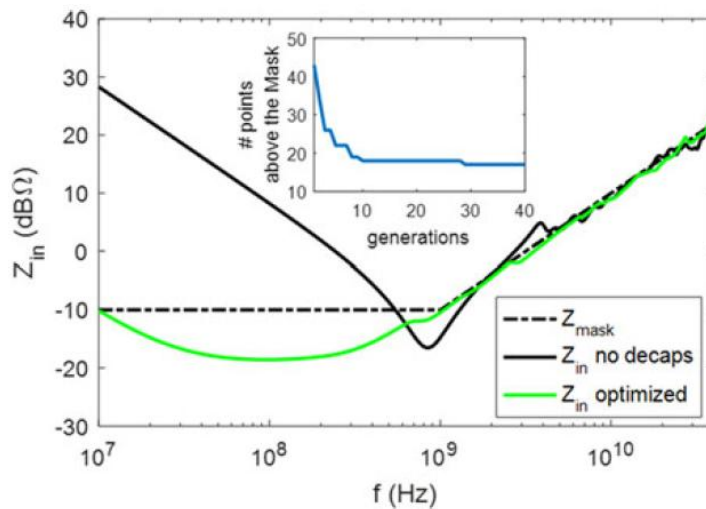
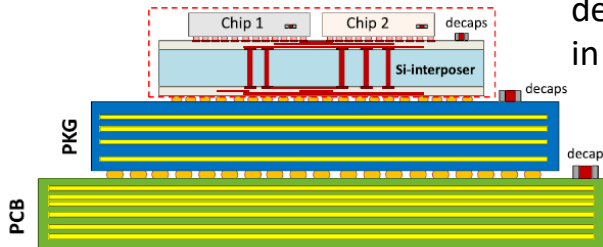
iter = 0



Metaheuristics for electronic design optimization

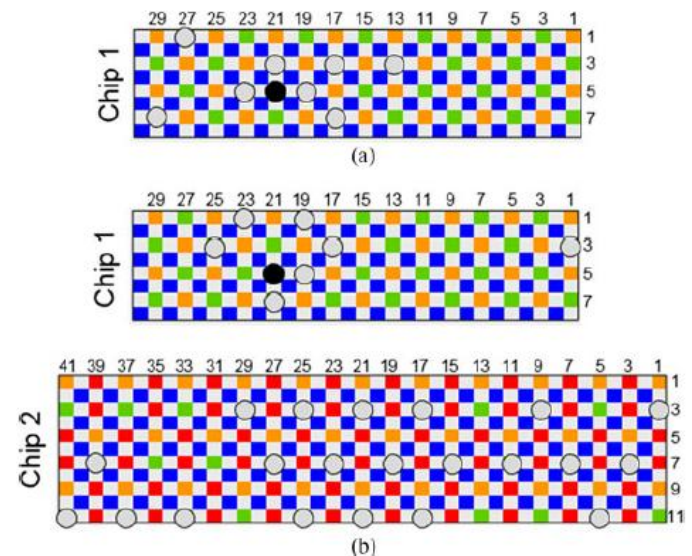
Genetic Algorithm/Particle Swarm Optimization - Example for electronic design

Optimal values/placement of decoupling capacitors (decaps) in multichip module ?



Possible positions of decaps on chips 1 and 2

Nature Inspired Algo.
(belong to GA)



Final positions of decaps on chips 1 and 2

S. Piersanti, F. de Paulis, C. Olivieri, A. Orlandi, "Decoupling Capacitors Placement for a Multichip PDN by a Nature-Inspired Algorithm", IEEE Trans. On EMC, vol. 60, no. 6, Dec. 2018.

Metaheuristics for electronic design optimization

► Multidisciplinary Optimization (MDO)

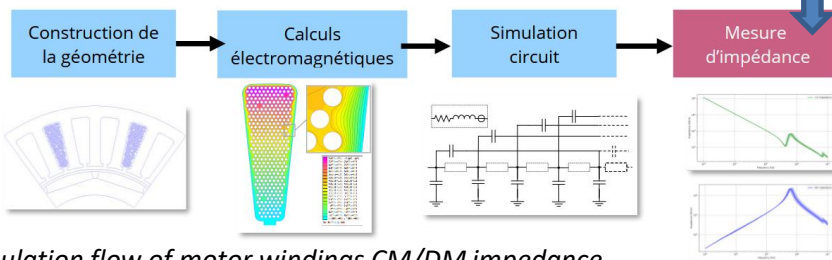
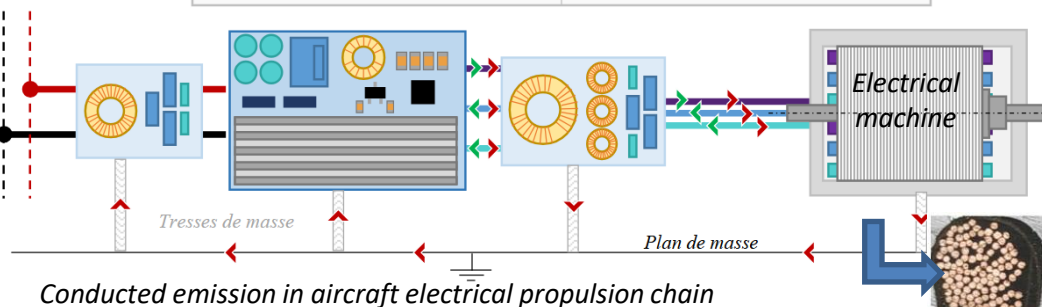
- ✓ Usage of optimization methods to solve design problems incorporating several disciplines simultaneously
- ✓ Examples of applications: in aircraft design: aero-structural optimization, electrical systems optimization
- ✓ Example: open source library for MDO:

GEMSEO

<https://gemseo.readthedocs.io>



► *Courant de Mode Commun* ► *Courant de Mode Différentiel*



Winding



Optimize geometry (conductor numbers, placement, insulator thickness...) to reduce EM emission, volume and weight

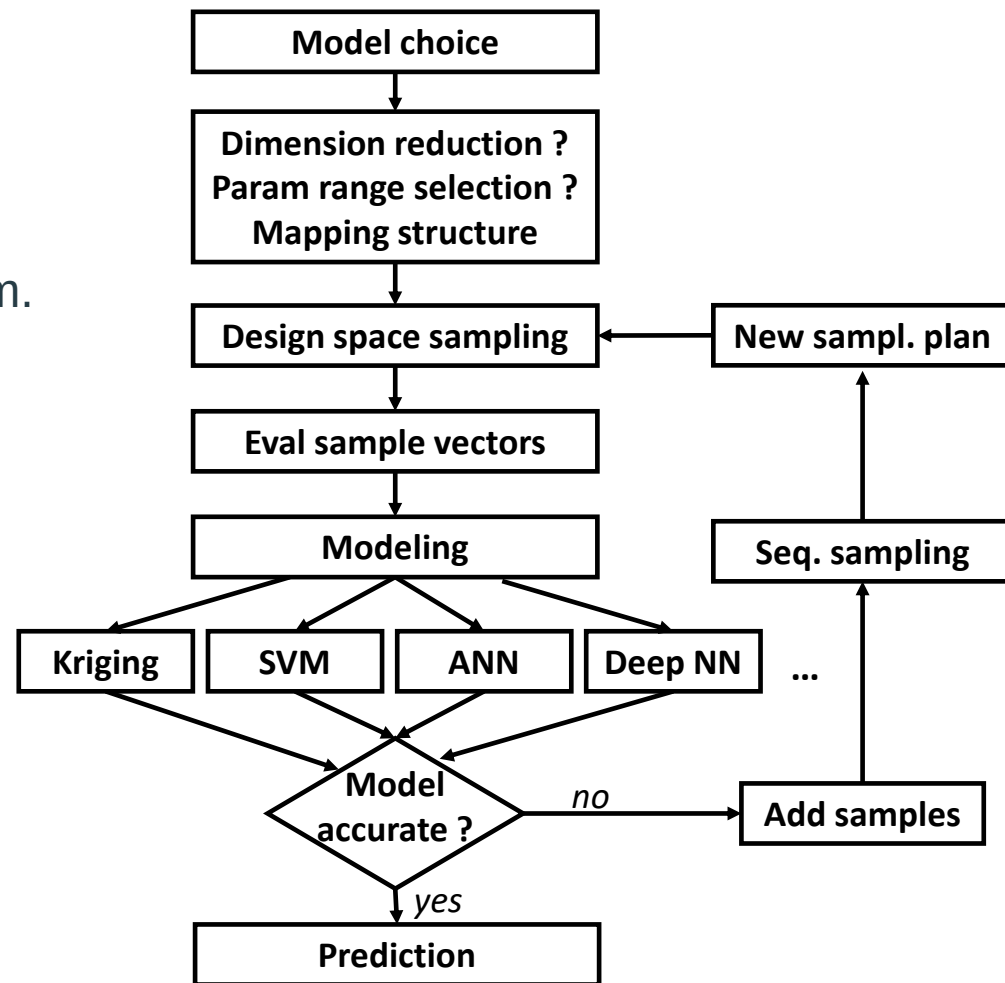
A. Piat, "Contribution à la prise en compte de paramètres incertains dans la modélisation prédictive des impédances de Mode Commun et Différentiel d'une machine électrique à grand nombre de spires", PhD Thesis, Univ. Paris-Saclay, 2024.

5. Supervised learning methods for surrogate modeling

Supervised learning methods for surrogate modeling

Supervised learning

- ✓ Construct surrogate black box or metamodels, i.e. an alternative mathematical model to simplify the representation of a complex system.
- ✓ Use given input data and corresponding labelled output data to train a model → accurate output predictions for new inputs.
- ✓ Advantages:
 - simplified representation
 - acceleration of simulation
 - protect IP
- ✓ Drawbacks:
 - no physics inside
 - no design details !!!

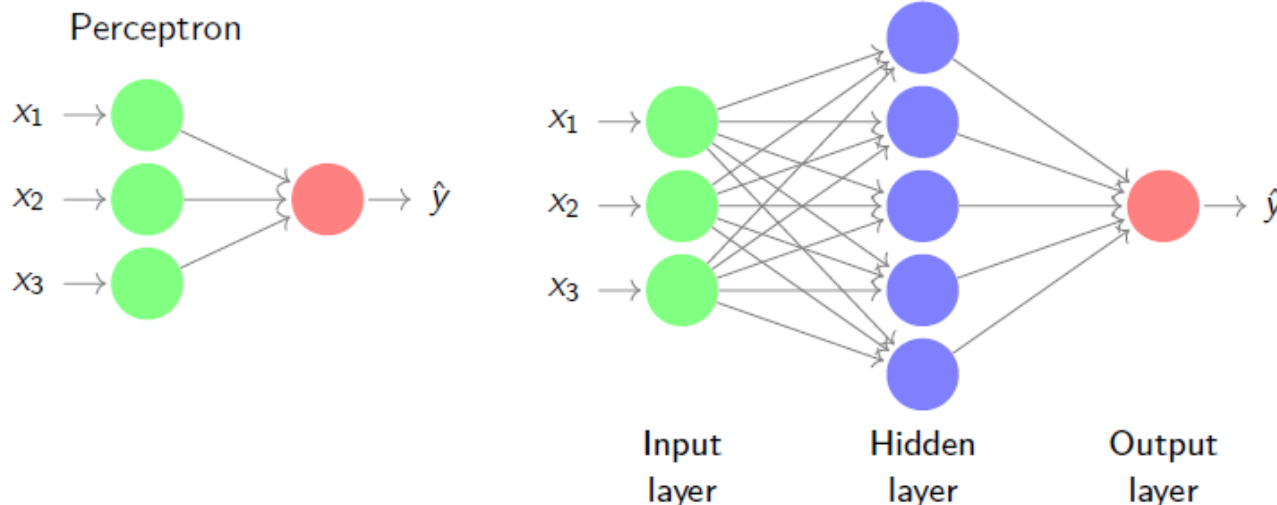


Generic flow for surrogate model construction

Supervised learning methods for surrogate modeling

► Artificial Neural Networks – Feedforward network

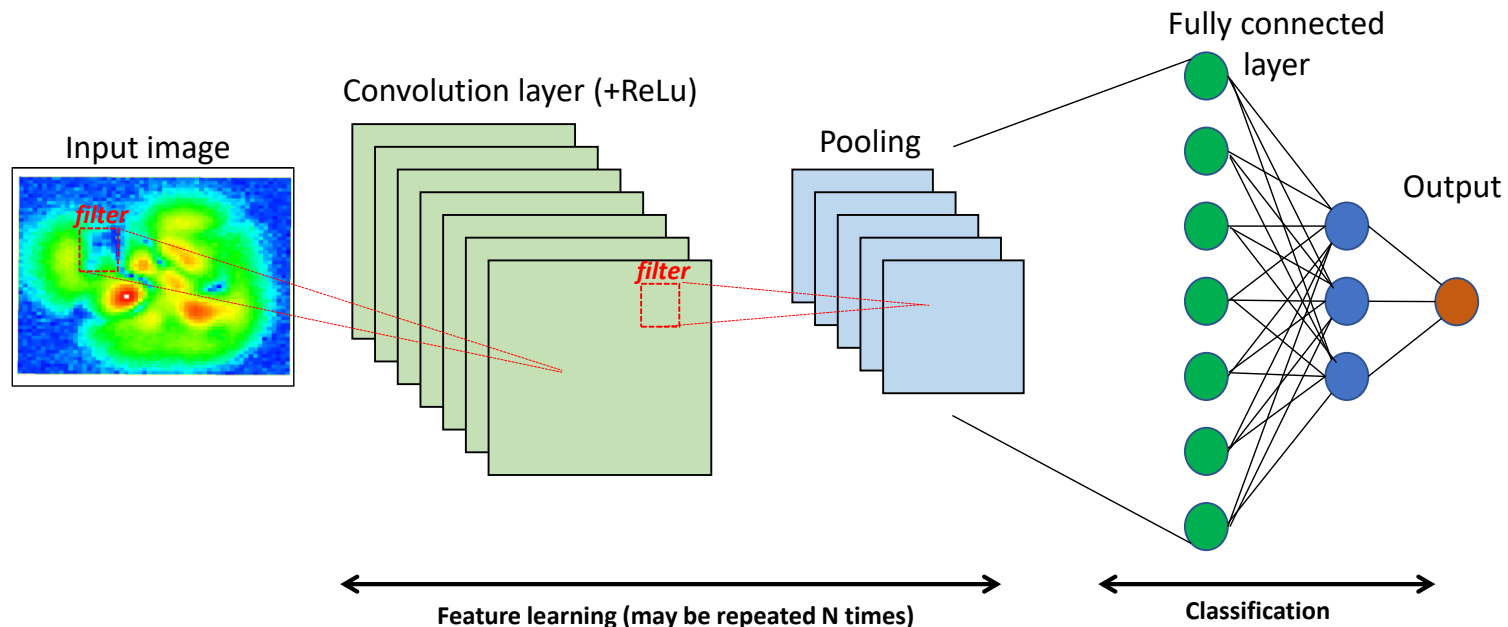
- ✓ Possibility to mimic any complex continuous function (Universal Approximation Theorem)
- ✓ Heterogeneous inputs can be considered
- ✓ High dimensions $\rightarrow$ multilayer perceptron
- ✓ Several layers = Deep neural network
- ✓ Risks: determination of the optimal hidden parameters (number of layer, weight, bias), risk of overfitting.



Supervised learning methods for surrogate modeling

► Artificial Neural Networks – FCNN, CNN and RNN

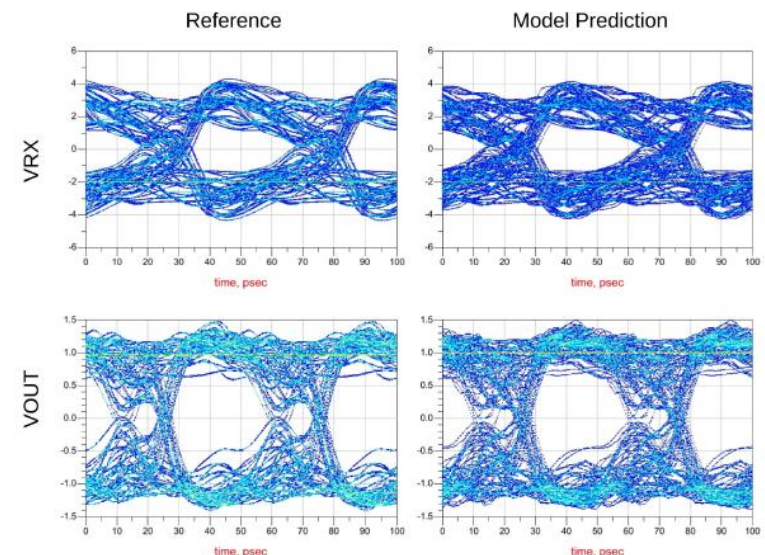
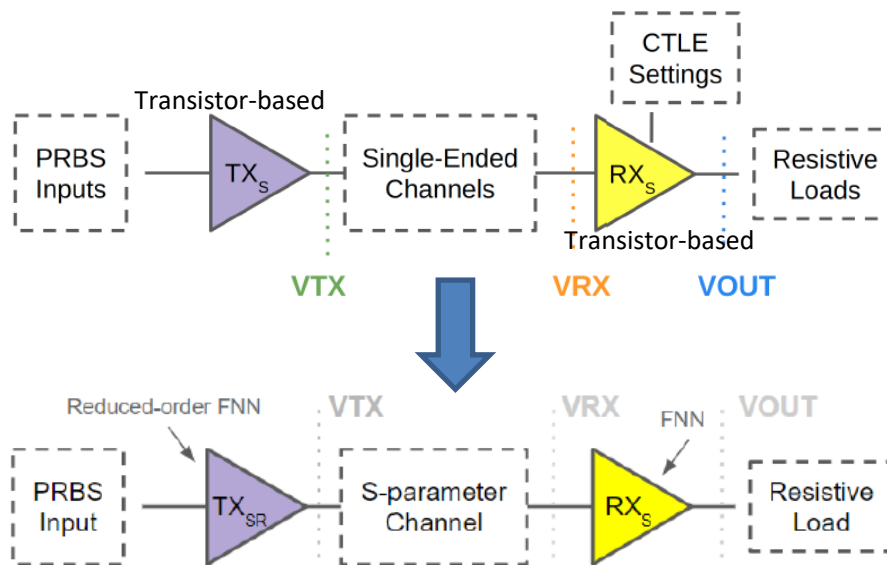
- ✓ Fully Connected Neural Network (FCNN): each input is connected to all the neurons. Unnecessary connections in high dimensions problems.
- ✓ Convolutional Neural Network (CNN): use of a filter (convolutional layer) to connect the previous to the next neuron layer → connection of each neuron to a local area of the input layer
- ✓ Recurrent Neural Network (RNN): connection of neurons of the same layer → capture dynamical effects of the modeled system (e.g. sequential circuits).



Supervised learning methods for surrogate modeling

▶ Artificial Neural Networks – Example for electronic design

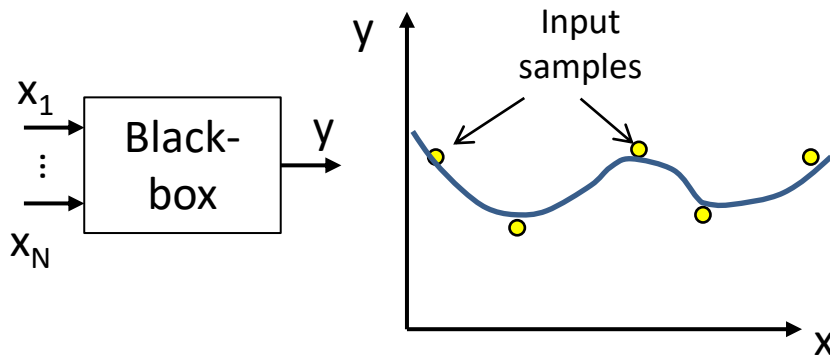
- Modular behavioral model of TX/RX buffers for rapid and accurate prediction of transient waveform in high-speed link (Signal Integrity prediction)
- Based on a 3 hidden layers feedforward neural network
- Trained using collection of discretized reference waveforms obtained under the typical operating conditions of the transceivers → extraction of kernel matrices independent of the channel.
- Alternative to IBIS file (table-based current sources and passive devices)
- 20x faster than conventional SPICE-like circuits.



Supervised learning methods for surrogate modeling

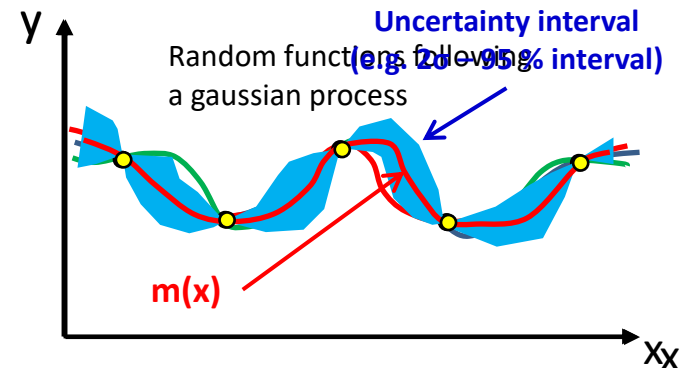
► Gaussian Process Regression (GPR) or kriging

Typical regression model



- ✓ Train one regression function from input data to minimize RMS error

GPR model



- ✓ Probabilistic supervised ML model
- ✓ Train a set of random functions (based on a given kernel k) following a multivariate gaussian distribution

- ✓ Evaluate the mean (estimated function) and variance (uncertainty) of the model:

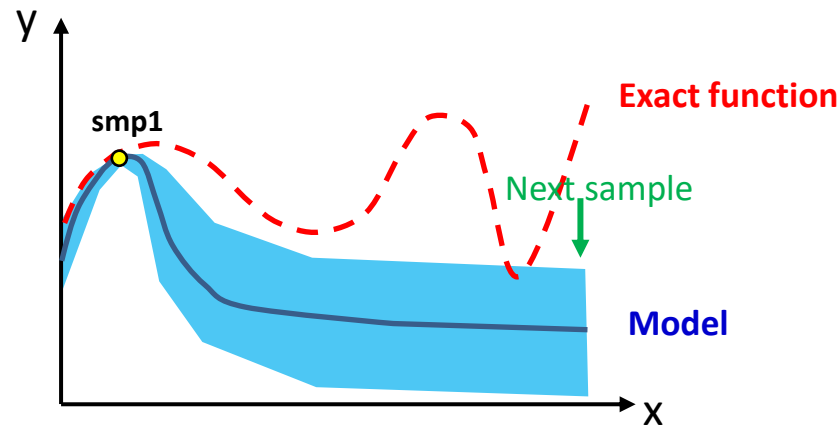
$$f(x) = GP(m(x), k(x, x'))$$

with
$$\begin{cases} m(x) = \mathbb{E}[f(x)] \\ k(x, x') = \mathbb{E}[(f(x) - m(x))(f(x') - m(x')))] \end{cases}$$

Supervised learning methods for surrogate modeling

► Bayesian optimization principle

Using GPR to sample iteratively and adaptively the design space in order to find the optimum value efficiently (e.g. in a case where the evaluation of the cost function is expensive).



- ✓ Exploration: increase the accuracy of the model → add points where the predicted uncertainty is maximum.
- ✓ To find the optimum → balance between exploration and exploitation.
- Require an **acquisition function** to determine the next sample (where it is minimized)

Example of acquisition function: lower confidence bound:

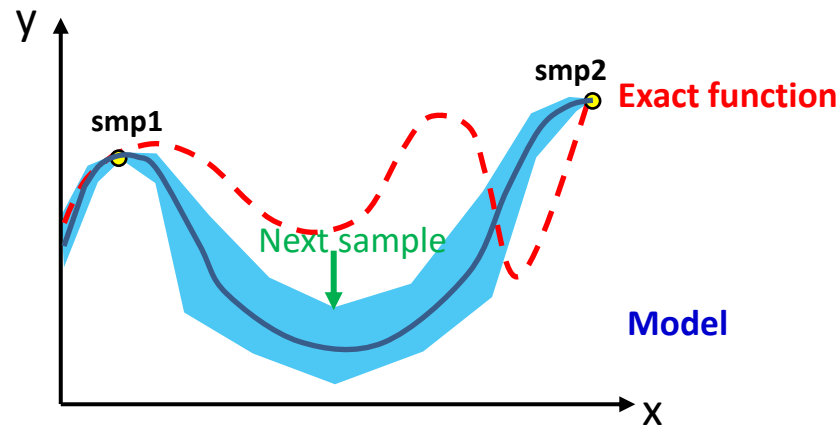
$$a(x; \lambda) = m(x) - \lambda \cdot \sigma(x)$$


 Increase λ improves exploration

Supervised learning methods for surrogate modeling

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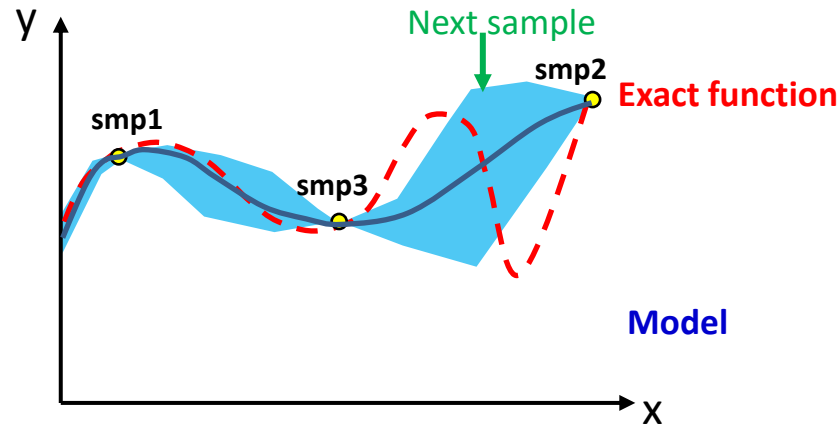
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↑
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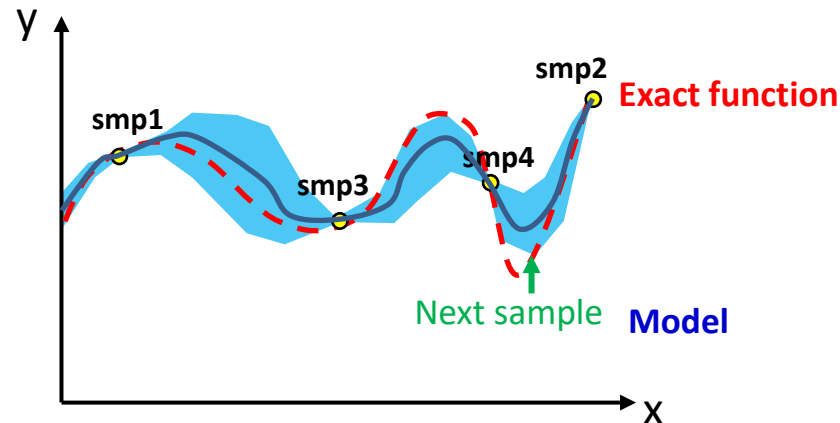
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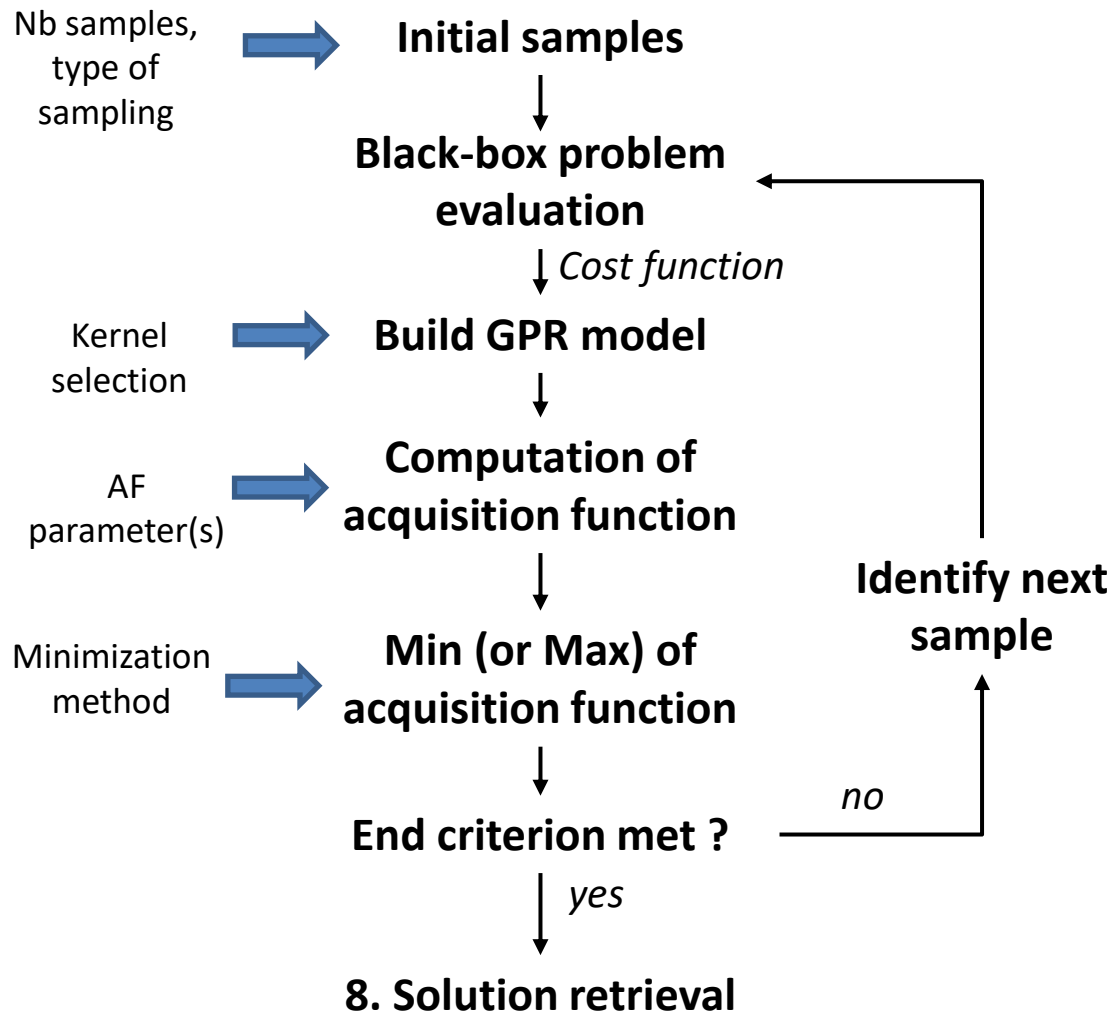
Example of acquisition function: lower confidence bound:

$$a(x; \lambda) = m(x) - \lambda \cdot \sigma(x)$$

↑
Increase λ improves exploration

Supervised learning methods for surrogate modeling

► Bayesian optimization – typical flow



✓ Advantages:

- Need few sample points to determine optimum
- → interesting when sample evaluation is costly
- Adaptive sampling
- Improvement of model at each iteration

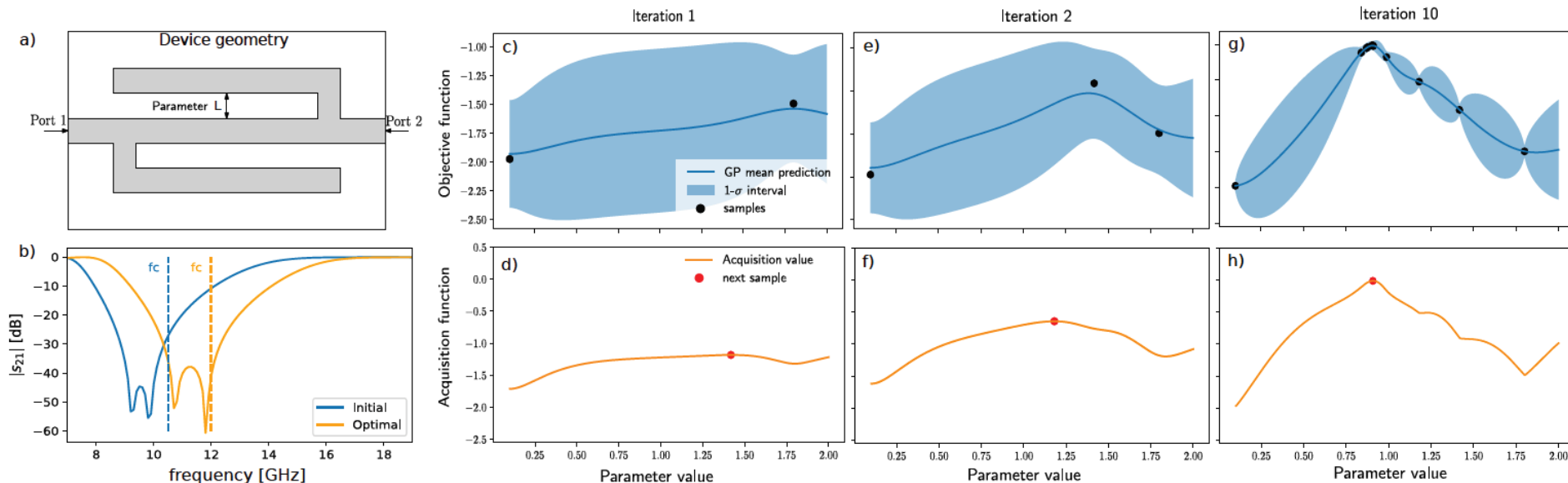
✓ Drawbacks:

- Dimensionality curse

Supervised learning methods for surrogate modeling

Bayesian optimization – Example for electronic design

- ✓ Optimization of a microstrip-based bandpass filter (folded stub filter)
- ✓ Find the best value for parameters L , S , h , ϵ to optimize the attenuation over the expected bandwidth (7 GHz) around the central frequency (12 GHz)
- ✓ BO algorithm controls a 3D full-wave EM simulator
- ✓ Around 30 samples are required



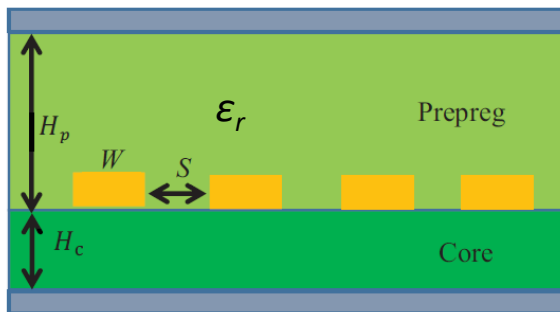
F. Garbuglia, J. Qing, N. Knudde, D. Spina, I. Couckuyt, D. Deschrijver, and T. Dhaene, "Bayesian active learning for multi-objective feasible region identification in microwave devices," *Electronics Letters*, vol. 57, no. 10, pp. 400–403, 2021

Supervised learning methods for surrogate modeling

► Bayesian optimization – Example for electronic design

- ✓ Coupling with ANN to accelerate the evaluation of the cost function
- ✓ Example: PCB/package stackup optimization:
 - Step 1: training of an ANN based on 2D EM simulation (352 samples)
 - Step 2 : Bayesian Optimization to minimize insertion loss and impedance mismatch (1000 iterations required).

Cost function: $f(\mathbf{x}) = |Z_c - Z_{target}| - a * Loss$



5 parameters to tune

Method	Time (minutes)
Genetic algo without NN	276
BO without NN	210
Genetic algo with NN	12
BO with NN	3.7

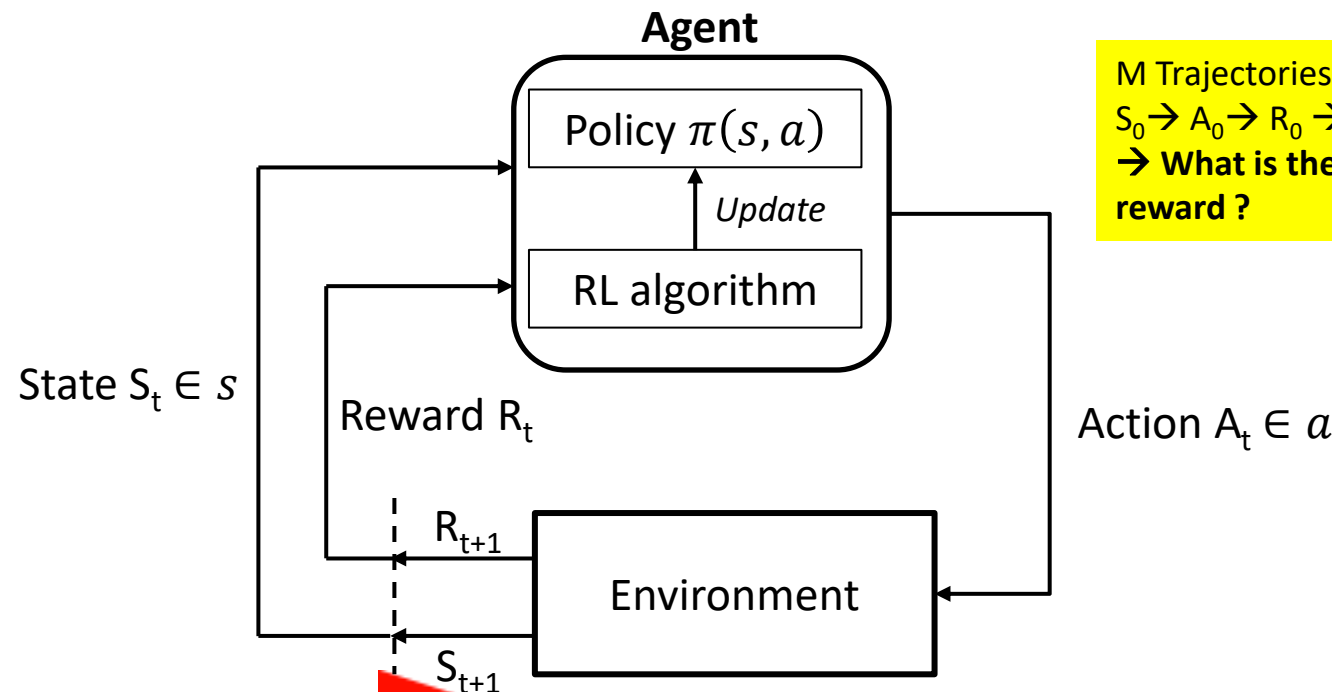
Z. Kiguradze, J. He, B. Mutnury, A. Chada, J. Drewniak, "Bayesian Optimization for Stack-up Design" IEEE Symp. on EMC, Signal and Power Integrity, Jul. 2019.

6. Reinforcement learning for electronic design optimization

Reinforcement learning for electronic design optimization

► Reinforcement learning (RL)

- ✓ Based on a Markov decision process (sequential decisions in an uncertain environment).
- ✓ RL trains an autonomous agent that iteratively learns from environment reaction with « no human in the loop » → Learning based on a Trial and error process
- ✓ The agent continuously interacts with the environment by observing the state, taking an action, obtaining a reward, observing the next state, and so on.
- ✓ Objective: Explore/exploit actions to maximize the gain (sum of rewards).



M Trajectories:

$S_0 \rightarrow A_0 \rightarrow R_0 \rightarrow S_1 \rightarrow A_1 \rightarrow R_1 \rightarrow \dots \rightarrow S_N \rightarrow A_N \rightarrow R_N$
→ **What is the policy that maximizes the total reward ?**

Reinforcement learning for electronic design optimization

► Example of RL algorithm: Q-learning

Initialization matrix $q(s,a) = 0$



Repeat M episodes



S_0 = initial state



While time step $t < N$



Choose action A_t from S_t according to policy (e.g. ϵ -greedy)



Apply A_t



Get S_{t+1} and R_{t+1} from environment



Update Q-function:

$$q(s,a) = (1 - \alpha)q(s,a) + \alpha \left(r + \gamma \max_{a'} q(s', a') \right)$$



End episodes ?

Increment t

r: reward to the agent
 α : learning factor [0;1]
 γ : actualisation factor [0;1]

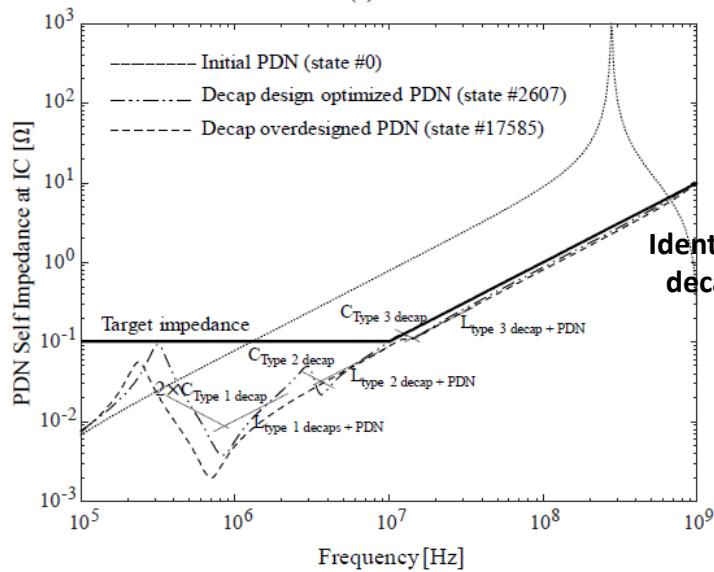
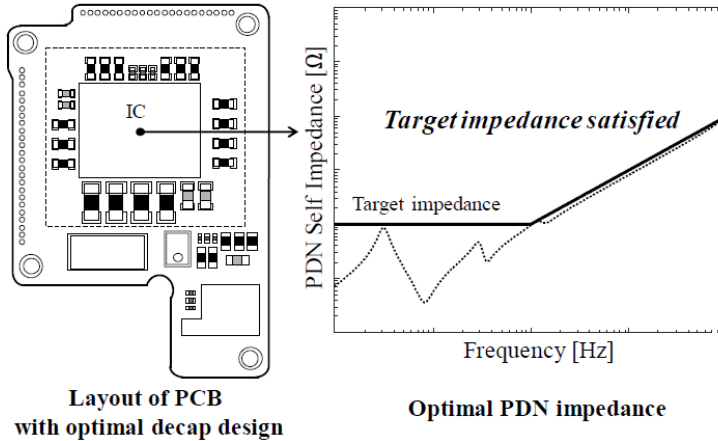
Best policy: $\pi^*(s,a) = \begin{cases} 1 & \text{if } a = \underset{a}{\operatorname{argmax}} q_\pi(a / s) \\ 0 & \end{cases}$



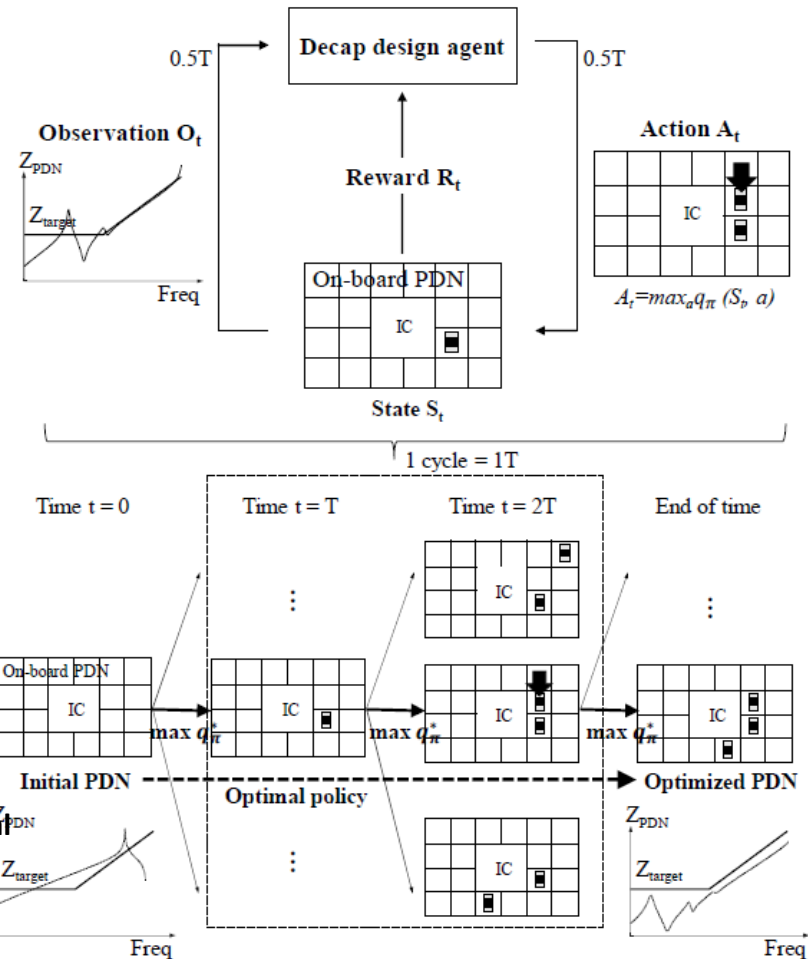
Reinforcement learning for electronic design optimization

Reinforcement Learning – Example for electronic design

Use RL to determine the optimal decoupling capacitors (decap) for power integrity purpose



Identification of 37 optimal decap designs (4 decaps)



H. Park, J. Park, S. Kim, D. Lho, S. Park, G. Park, K. Cho, J. Kim, "Reinforcement Learning-based Optimal On-board Decoupling Capacitor Design Method," 27th Conf. On Electrical Perf. Of Electronic Packaging and Systems, Oct. 2018.

Some references

- ✓ INSA course I4AEML21 – Machine Learning, <https://moodle.insa-toulouse.fr/course/view.php?id=1790>
- ✓ H. Ren, J. Hu, « Machine Learning Applications in Electronic Design Automation », Springer, 2022, ISBN 978-3-031-13073-1.
- ✓ M. J. Kochenderfer, T. A. Wheeler « Algorithms for Optimization – 2nd Edition », MIT Press, 2025.
- ✓ L. Jiang, « Machine Learning for EMC/SI/PI – Blackbox, Physics Recovery, and Decision Making », IEEE EMC Magazine, vol. 12, Quarter 4, 2023.
- ✓ F. Garbuglia, D. Deschrijver, T. Dhaene, « On the Role of Bayesian Learning for Electronic Design Automation: A Survey », IEEE EMC Magazine, vol. 12, Quarter 4, 2023.
- ✓ M. Swaminathan, H. M. Torun, H. Yu, J. A. Hejase, W. D. Becker, «Demystifying Machine Learning for Signal and Power Integrity Problems in Packaging», IEEE Trans. On Components, Packaging and Manuf. Techno, vol. 10, no. 8, Aug. 2020.
- ✓ B. Shahriari, K. Swersky, Z. Wang, R. P. Adams, N. de Freitas, “Taking the Human Out of the Loop: A Review of Bayesian Optimization”, Proc. of the IEEE, vol. 104, No. 1, Jan. 2016.